Routing protocols for wireless ad hoc Networks

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Outline

* Wireless Networks: Classifications
* Challenges: Ad-hoc Network & Design issues
* Adhoc Routing: Requirements
* Adhoc Routing: Classification
* Reactive Roting
* Pro-active Routing
* Hybrid Routing

Wireless Networks

* Need: Access computing and communication services, on the move

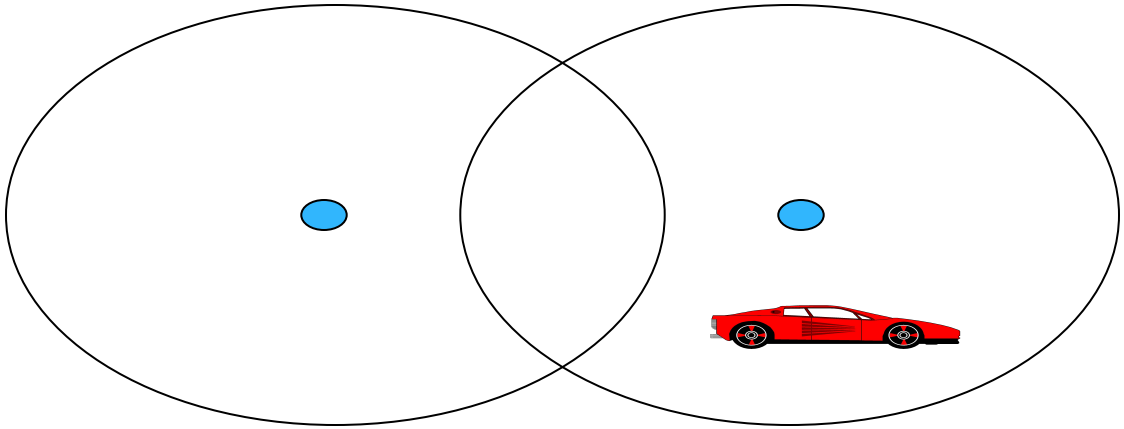
* Infrastructure-based Networks
  * traditional cellular systems (base station infrastructure)

* Wireless LANs
  * Infrared (IrDA) or radio links (Wavelan)
  * very flexible within the reception area; ad-hoc networks possible
  * low bandwidth compared to wired networks (1-10 Mbit/s)

* Ad hoc Networks
  * useful when infrastructure not available, impractical, or expensive
  * military applications, rescue, home networking

Cellular Networks

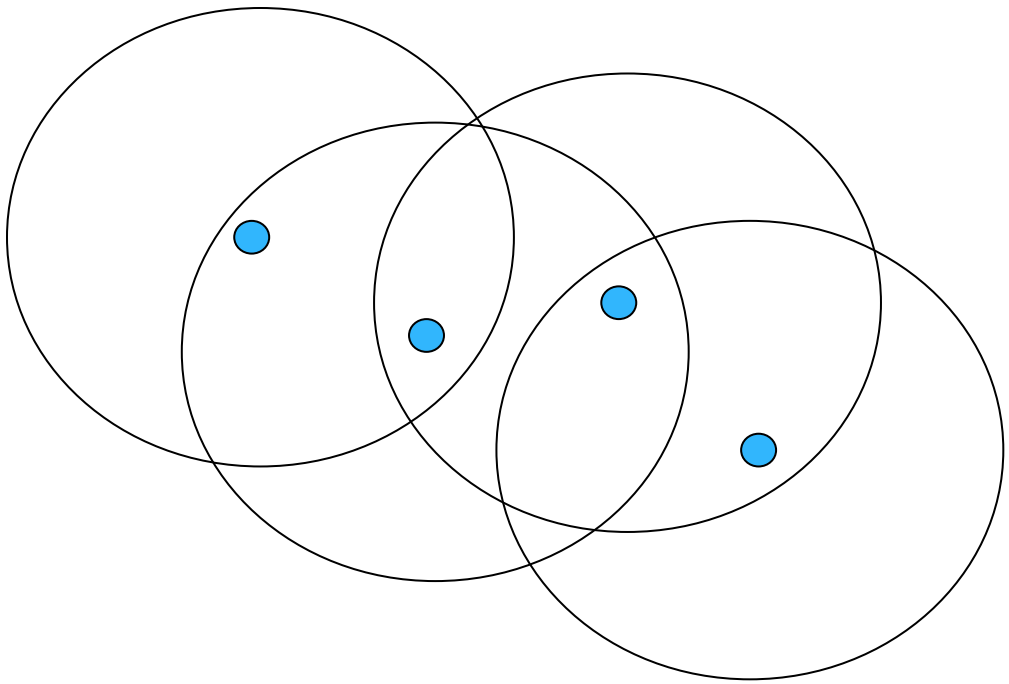
* Single hop wireless connectivity to the wired world
  * Space divided into cells
  * A base station is responsible to communicate with hosts in its cell
  * Mobile hosts can change cells while communicating
  * Hand-off occurs when a mobile host starts communicating via a new base station

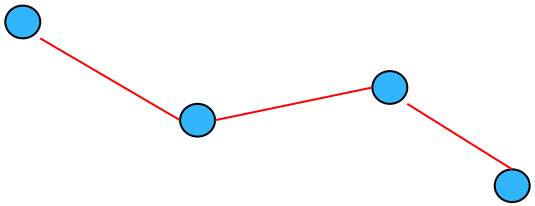


Multi-hop Wireless

May need to traverse multiple links to reach destination

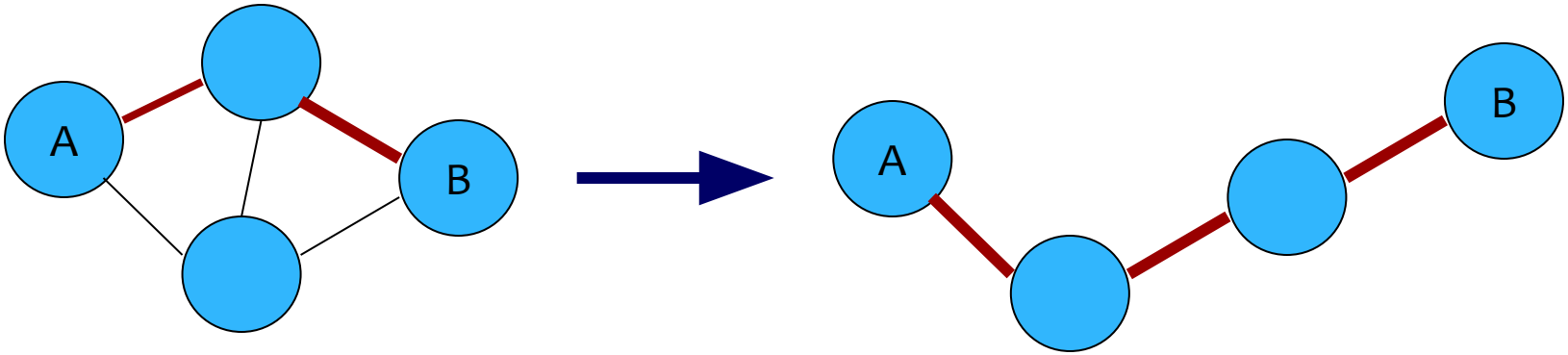
Mobility causes route change





Ad-hoc Networks

* Host movement frequent
* Topology change frequent



* No cellular infrastructure. Multi-hop wireless links.
* Data must be routed via intermediate nodes.

Why Ad Hoc Networks ?

* Setting up of fixed access points and backbone infrastructure is not always viable
  * Infrastructure may not be present in a disaster area or war zone
  * Infrastructure may not be practical for short-range radios; Bluetooth (range ~ 10m)

* Ad hoc networks:
  * Do not need backbone infrastructure support
  * Easy to deploy
  * Useful when infrastructure is absent, destroyed or impractical

Applications

* Personal area networking
  * cell phone, laptop, ear phone, wrist watch
* Military environments
  * soldiers, tanks, planes
* Civilian environments
  * taxi cab network
  * meeting rooms
  * sports stadiums
  * boats, small aircraft
* Emergency operations
  * search-and-rescue
  * policing and fire fighting

Challenges

- Limitations of the Wireless Network
  - packet loss due to transmission errors
  - variable capacity links
  - frequent disconnections/partitions
  - limited communication bandwidth
  - Broadcast nature of the communications

- Limitations Imposed by Mobility
  - dynamically changing topologies/routes
  - lack of mobility awareness by system/applications

- Limitations of the Mobile Computer
  - short battery lifetime
  - limited capacities

Adhoc Routing: Requirements

- Fully distributed, no critical nodes
- Allow for random node events (e.g. entering, leaving, neighbor changes)
- Minimum delay to determine path (at transmission time)
- Minimize storage requirements
- Must remove (or not propagate) invalid paths

Unicast Routing

* Many protocols have been proposed

* Some specifically invented for MANET
* Others adapted from protocols for wired networks

* No single protocol works well in all environments
  * some attempts made to develop adaptive/hybrid protocols

Classifications: Routing

* Proactive protocols
  * Traditional distributed shortest-path protocols
  * Maintain routes between every host pair at all times
  * Based on periodic updates; High routing overhead
  * Example: DSDV (Destination Sequenced Distance Vector)

* Reactive protocols
  * Determine route if and when needed
  * Source initiates route discovery
  * Example: DSR (Dynamic Source Routing)

* Hybrid protocols
  * Adaptive; Combination of proactive and reactive
  * Example : ZRP (Zone Routing Protocol)

Routing: Features

* Proactive protocols
  * Always maintain routes
  * Little or no delay for route determination
  * Consume bandwidth to keep routes up-to-date
  * Maintain routes which may never be used
* Reactive protocols
  * Lower overhead since routes are determined on demand
  * Significant delay in route determination
  * Employ flooding (global search)
  * Control traffic may be bursty
* Which approach achieves a better trade-off?
  * It depends on the traffic and mobility patterns

Ad hoc routing protocols – classification

* ***When***?

* Option 1: Routing protocol ***always*** tries to keep its routing data up-to-date
  * Protocol is ***proactive*** (active before tables are actually needed) or ***table-driven***

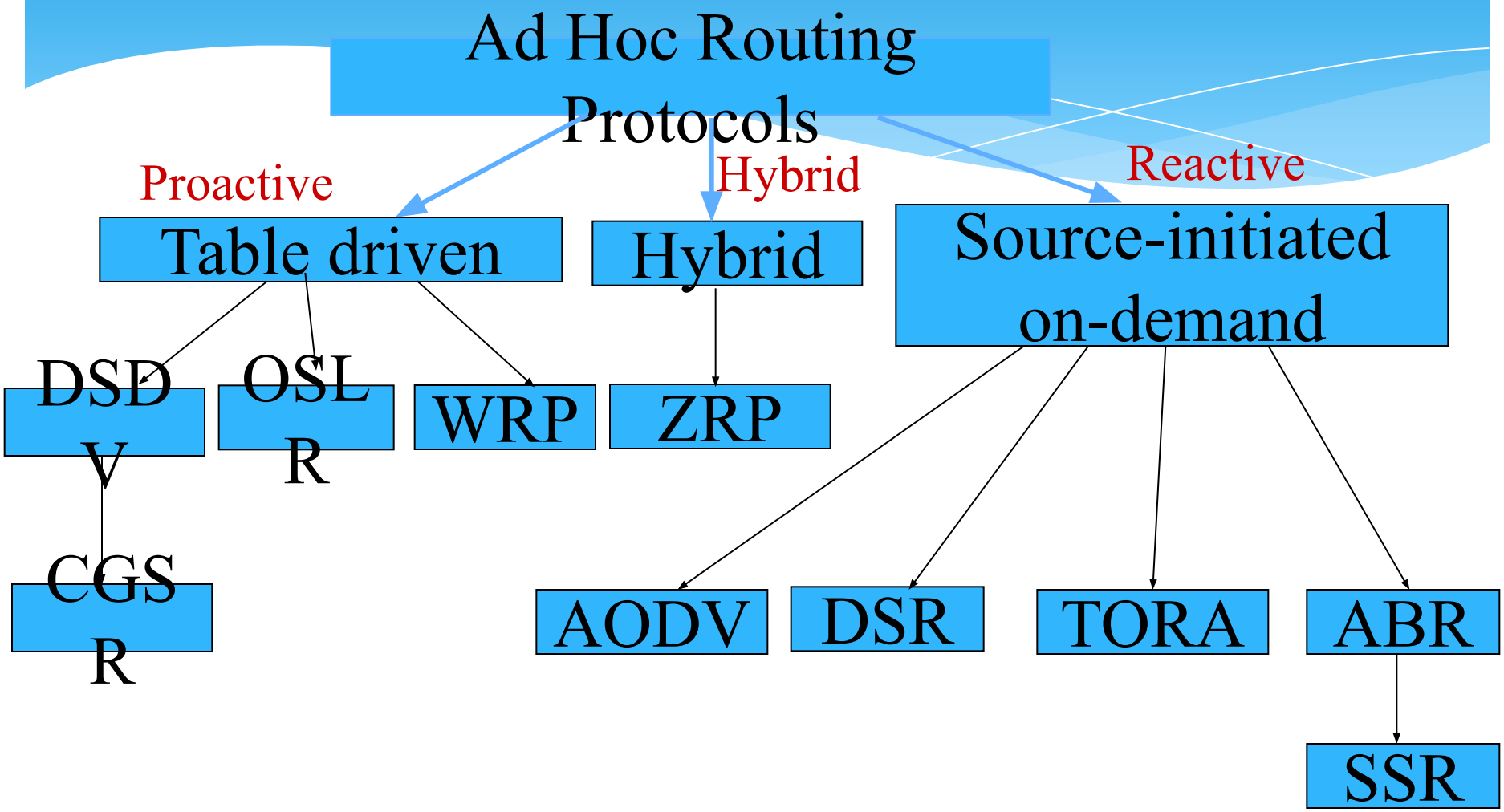
* Option 2: Route is only determined when actually needed
  * Protocol operates ***on demand***

* Option 3: Combine these behaviors
  * ***Hybrid*** protocols

Ad hoc routing protocols – classification

* Network Structure : flat or hierarchical?
* – Flat addressing scheme similar to IEEE 802.3 LANs
* – Globally unique addressing mechanism for nodes
  * Compare topology control, traditional routing
* Which data is used to identify nodes?
  * An arbitrary identifier?
  * The ***position*** of a node?
    * ***geographic*** routing protocols because choice of next hop neighbor can be computed based on destination address
  * Identifiers that are not arbitrary, but carry some structure?
    * As in traditional routing
    * Structure akin to position, on a logical level?

Classification (cont.)



Dynamic Source Routing (DSR)

* Cooperative nodes
* Relatively small network diameter (5-10 hops)
* Detectable packet error
* Unidirectional or bidirectional link
* Promiscuous mode (optional)

DSR : Reactive Protocol

* In a reactive protocol, how to forward a packet to destination?
  * Initially, no information about next hop is available at all
  * One (only?) possible recourse: Send packet to ***all*** neighbors – flood the network
  * Hope: At some point, packet will reach destination and an answer is sent back – use this answer for ***backward learning*** the route from destination to source

* Practically: ***Dynamic Source Routing (DSR)***
  * Use separate ***route request/route reply*** packets to discover route
    * Data packets only sent once route has been established
    * Discovery packets smaller than data packets
  * Store routing information in the discovery packets

Phases

* Route discovery
  * When node S wants to send a packet to node D, but does not know a route to D, node S initiates a route discovery

* Route maintenance
* Mechanisms “on-demand”
  * No periodic routing advertisement
  * No link status sensing
  * No neighbor detection packets
* Routes caching

DSR

* Source node S floods Route Request (RREQ) to its' neighbors
* Each node *appends own identifier in the header* when forwarding RREQ and floods the message
* The destination responds with a Route Reply message containing the path

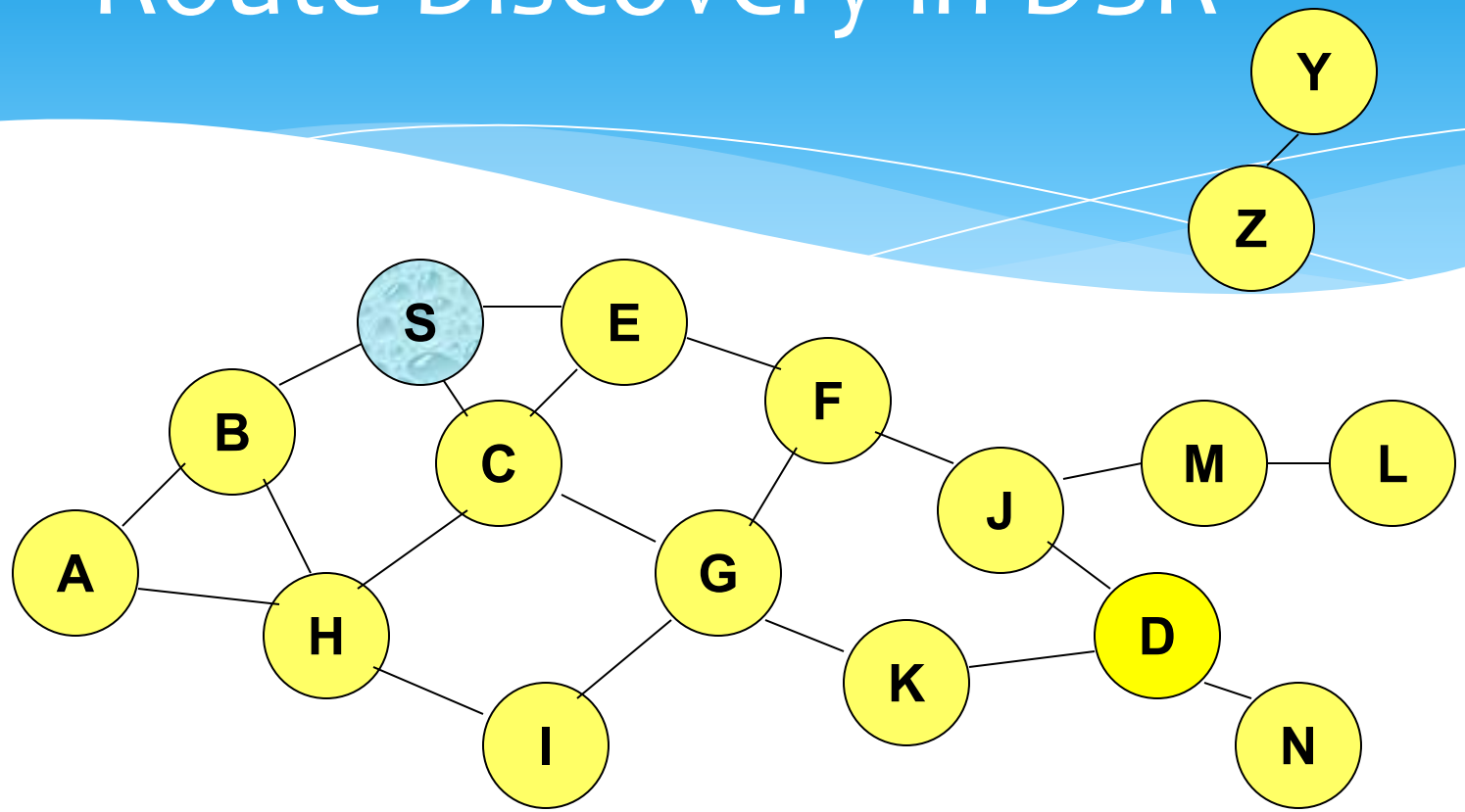
Dynamic Source Routing (DSR)

* Node S on receiving RREP, caches the route included in the RREP
* When node S sends a data packet to D, the entire route is included in the packet header
  * hence the name source routing
* Intermediate nodes use the source route included in a packet to determine to whom a packet should be forwarded

DSR

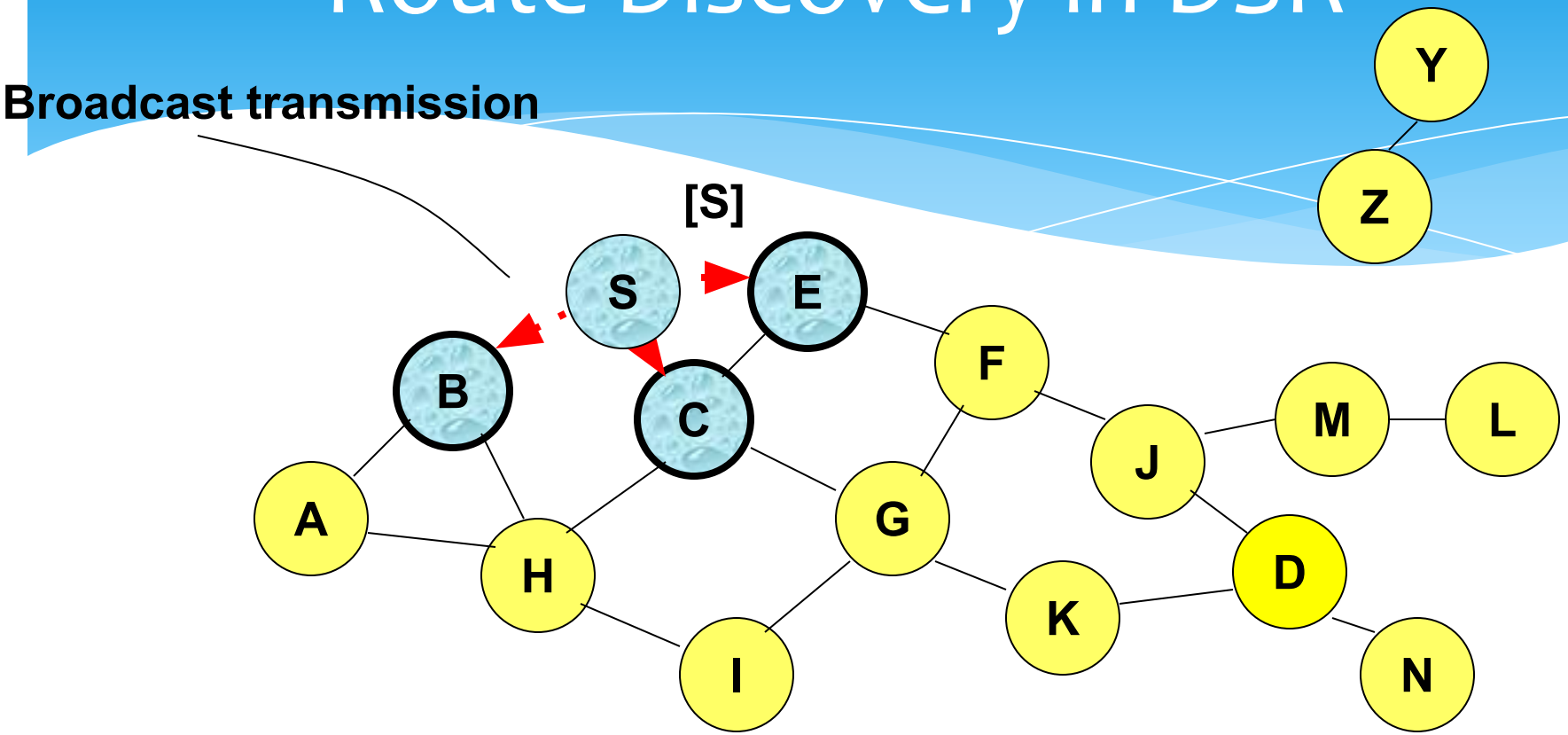
* A node can only flood the RouteRequestpacket if it has not already flooded it
* On link failure:
  * Adjacent node sends a RouteErrormessage to the source
  * The source will remove the route from its route entry list

Route Discovery in DSR



Represents a node that has received RREQ for D from S

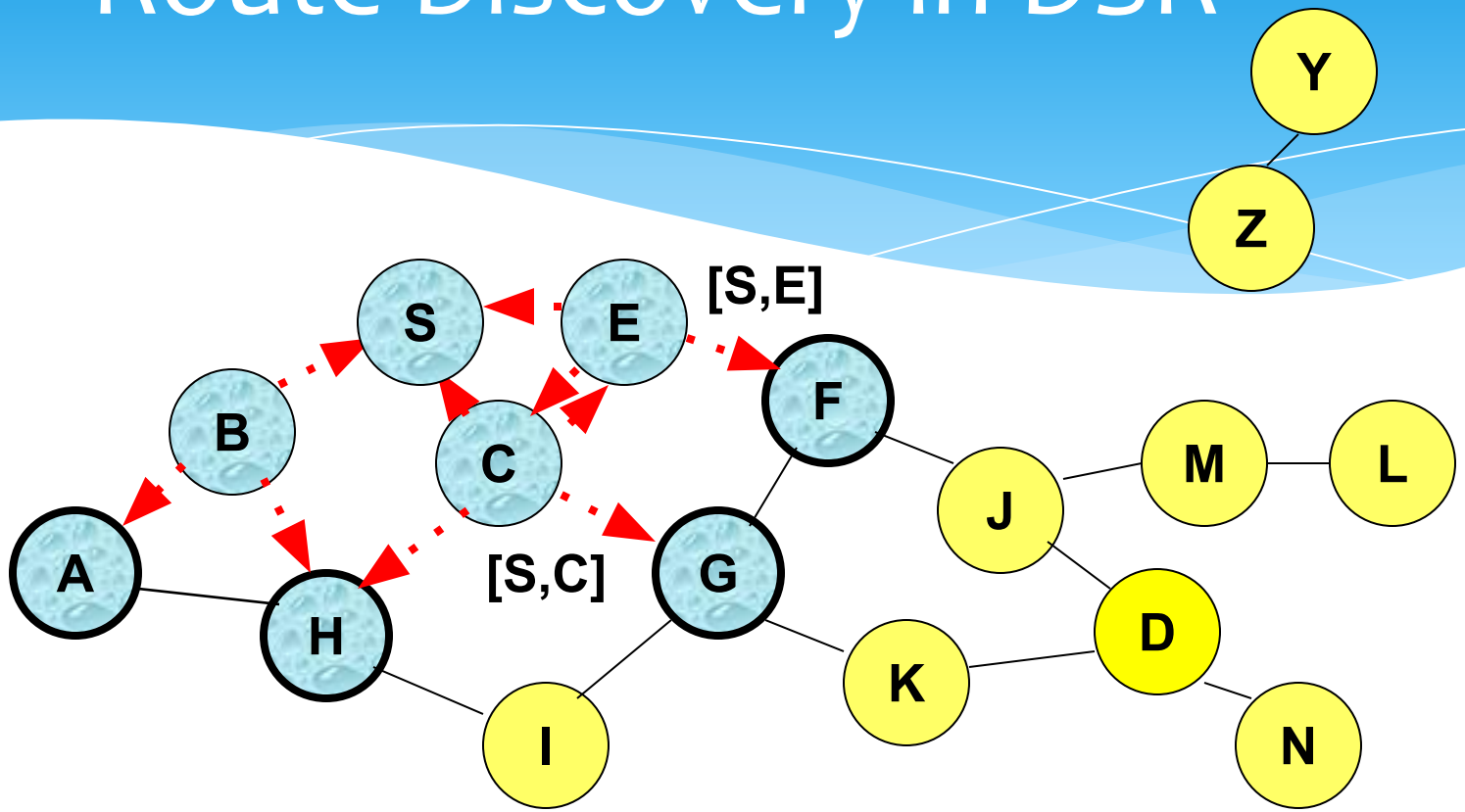
Route Discovery in DSR



Represents transmission of RREQ

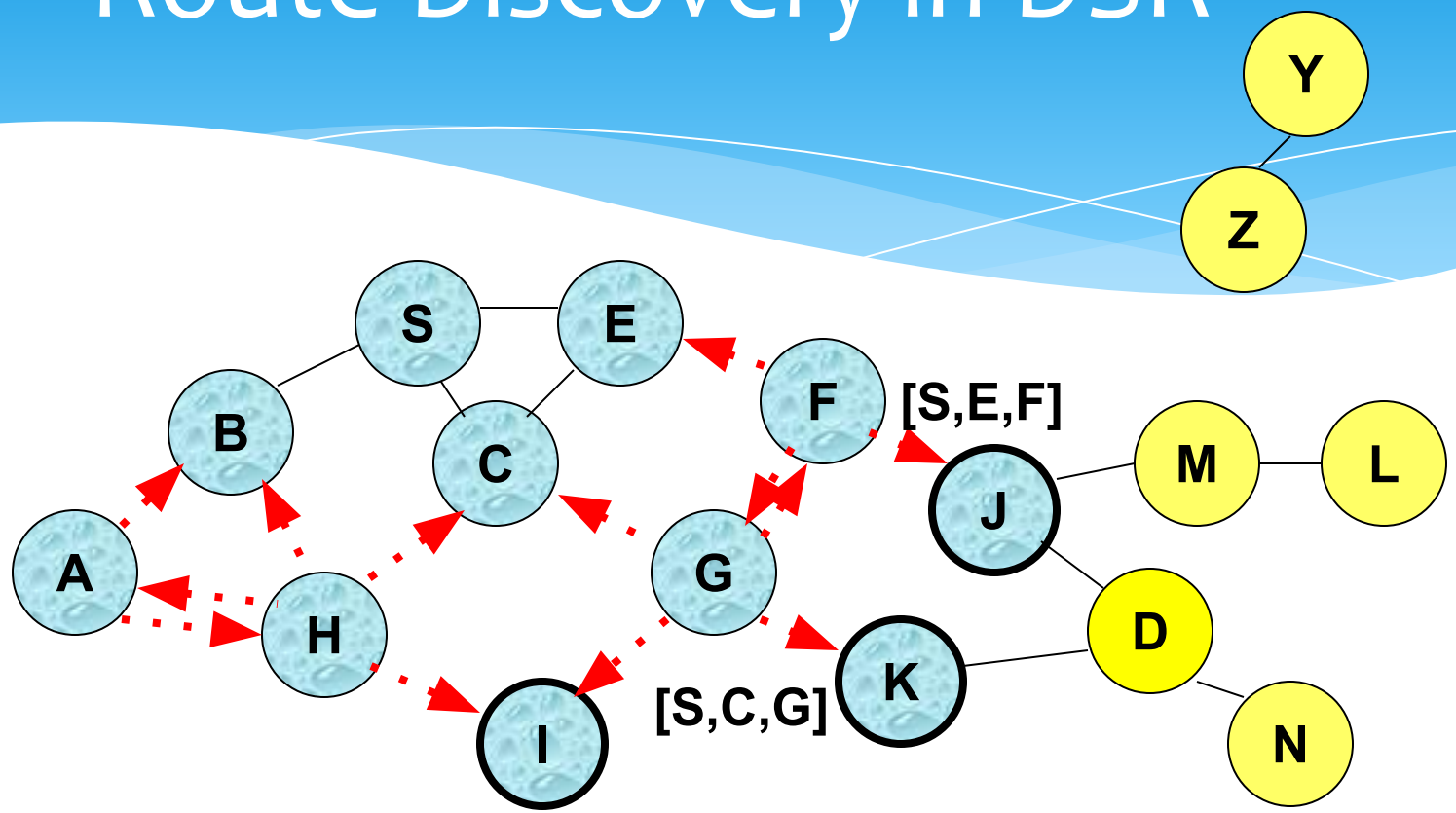
[X,Y] Represents list of identifiers appended to RREQ

Route Discovery in DSR



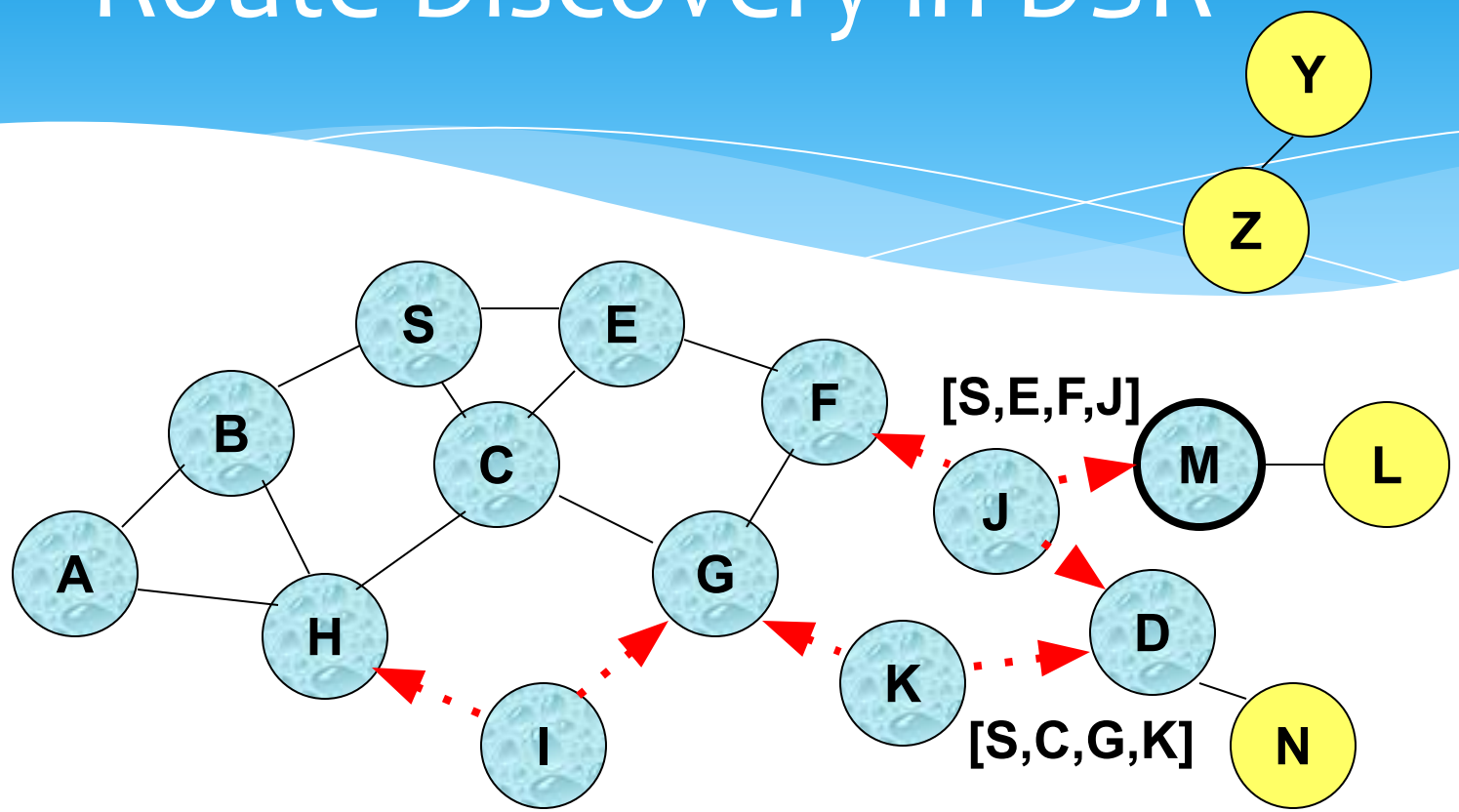
- **Node H receives packet RREQ from two neighbors:**
  potential for collision

Route Discovery in DSR



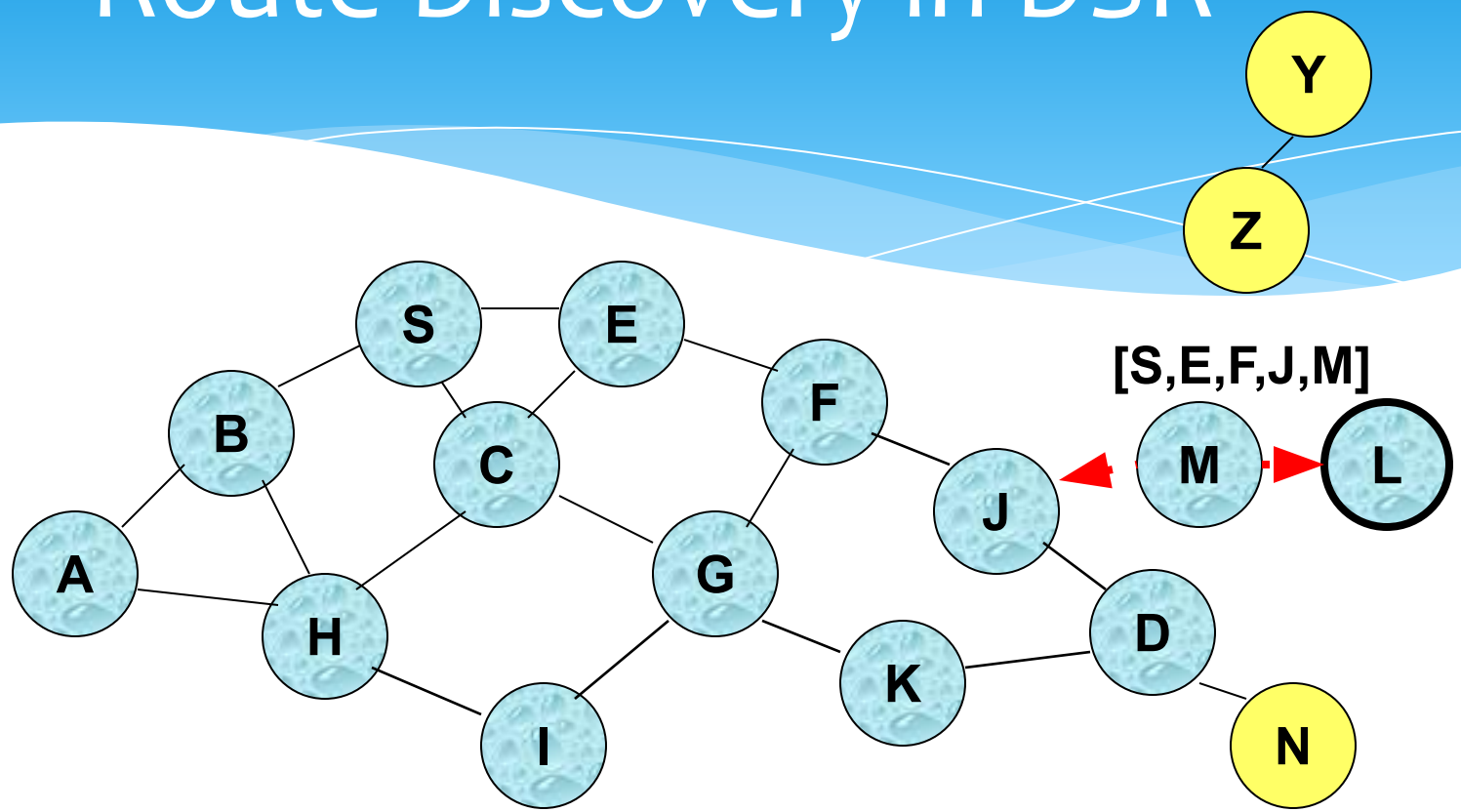
- Node C receives RREQ from G and H, but does not forward it again, because node C has **already forwarded RREQ** once

Route Discovery in DSR



- Nodes J and K both broadcast RREQ to node D
- Since nodes J and K are hidden from each other, their transmissions may collide

Route Discovery in DSR

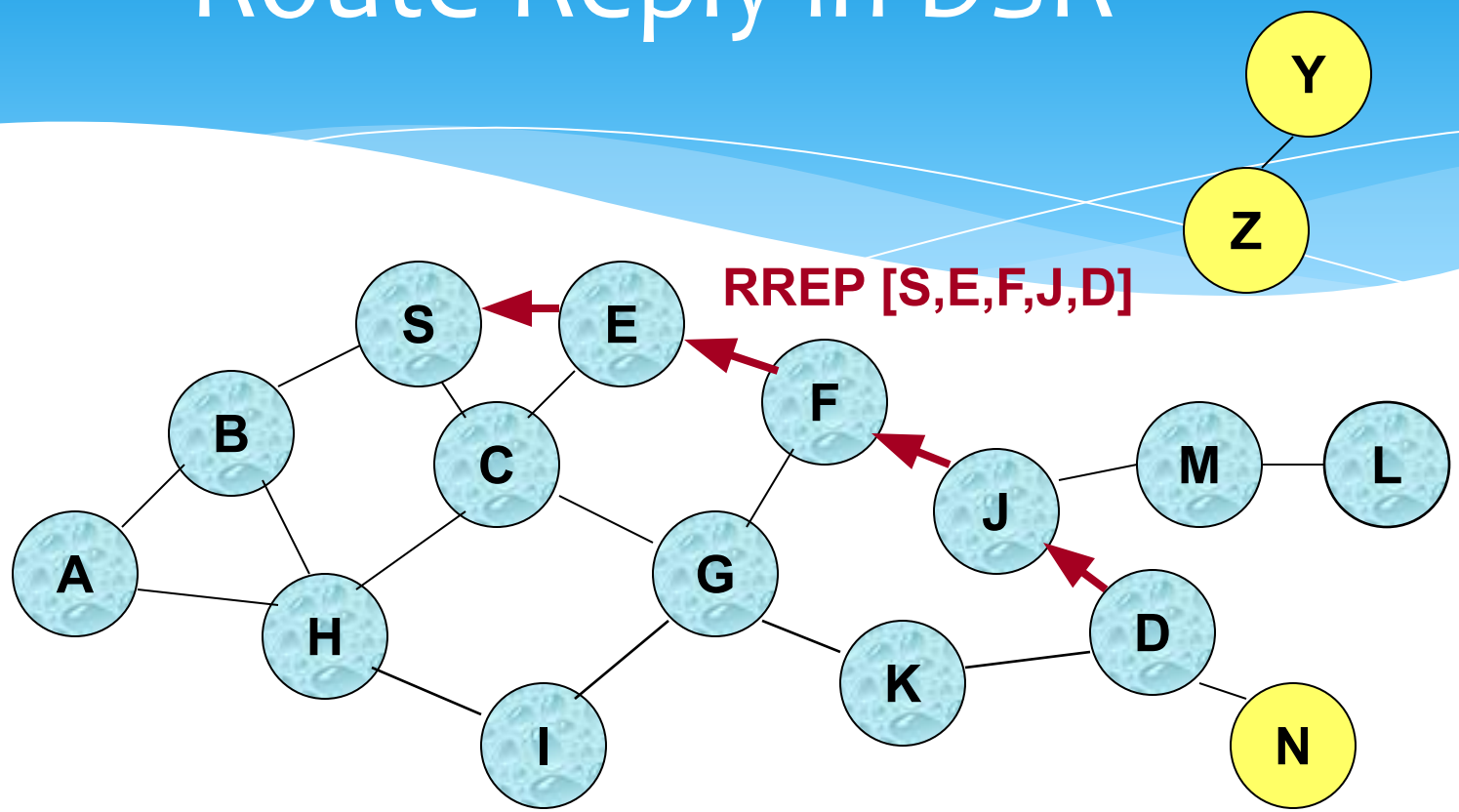


- Node D **does not forward** RREQ, because node D is the **intended target** of the route discovery

Route Discovery in DSR

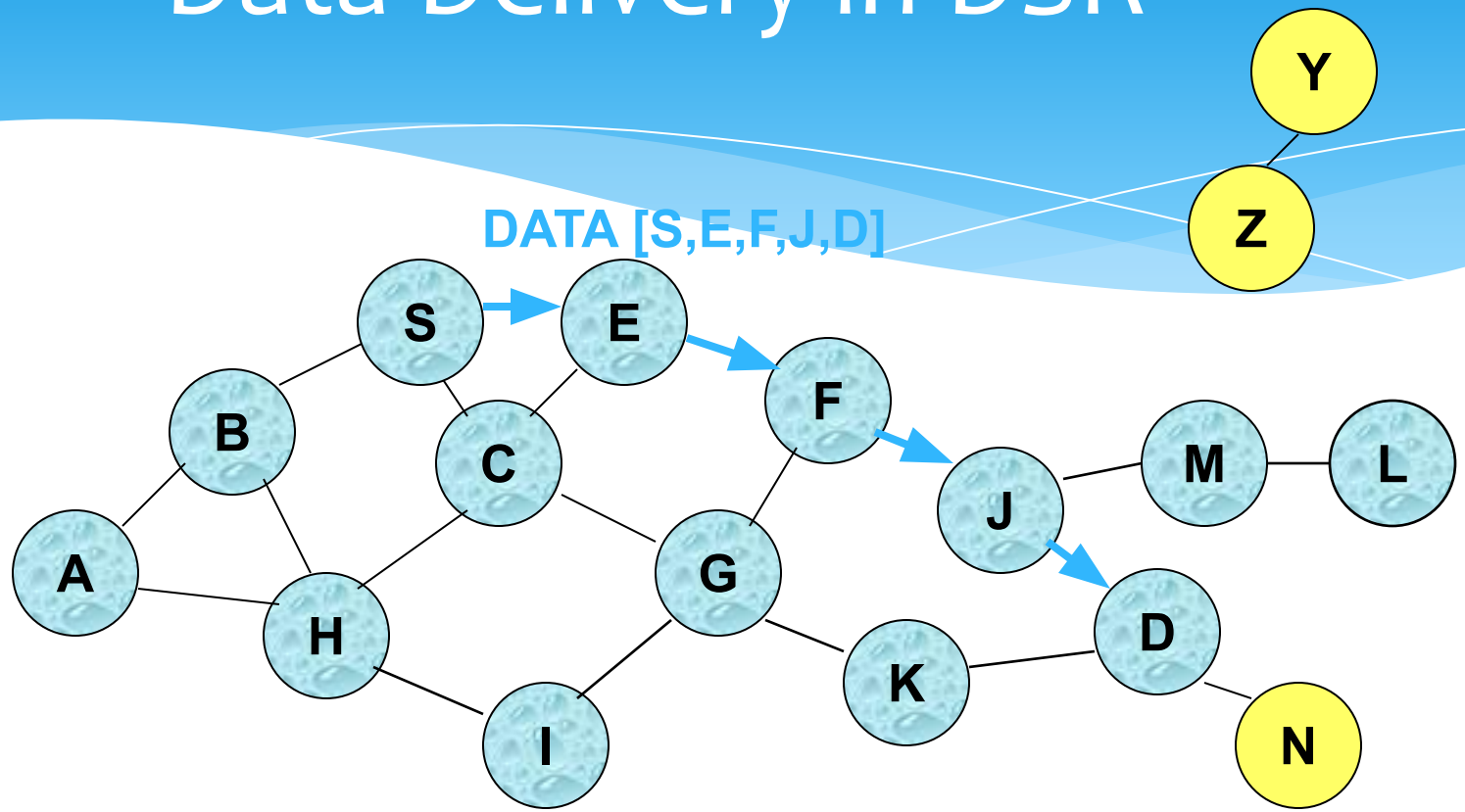
* Destination D on receiving the first RREQ, sends a Route Reply (RREP)
* RREP is sent on a route obtained by reversing the route appended to received RREQ
* RREP includes the route from S to D on which RREQ was received by node D

Route Reply in DSR



Represents RREP control message

Data Delivery in DSR



Packet header size grows with route length

DSR Optimization: Route Caching

* Each node caches a new route it learns by *any means*
* When node S finds route [S,E,F,J,D] to node D, node S also learns route [S,E,F] to node F
* When node K receives Route Request [S,C,G] destined for node, node K learns route [K,G,C,S] to node S
* When node F forwards Route Reply RREP [S,E,F,J,D], node F learns route [F,J,D] to node D
* When node E forwards Data [S,E,F,J,D] it learns route [E,F,J,D] to node D
* A node may also learn a route when it overhears Data
* Problem: Stale caches may increase overheads

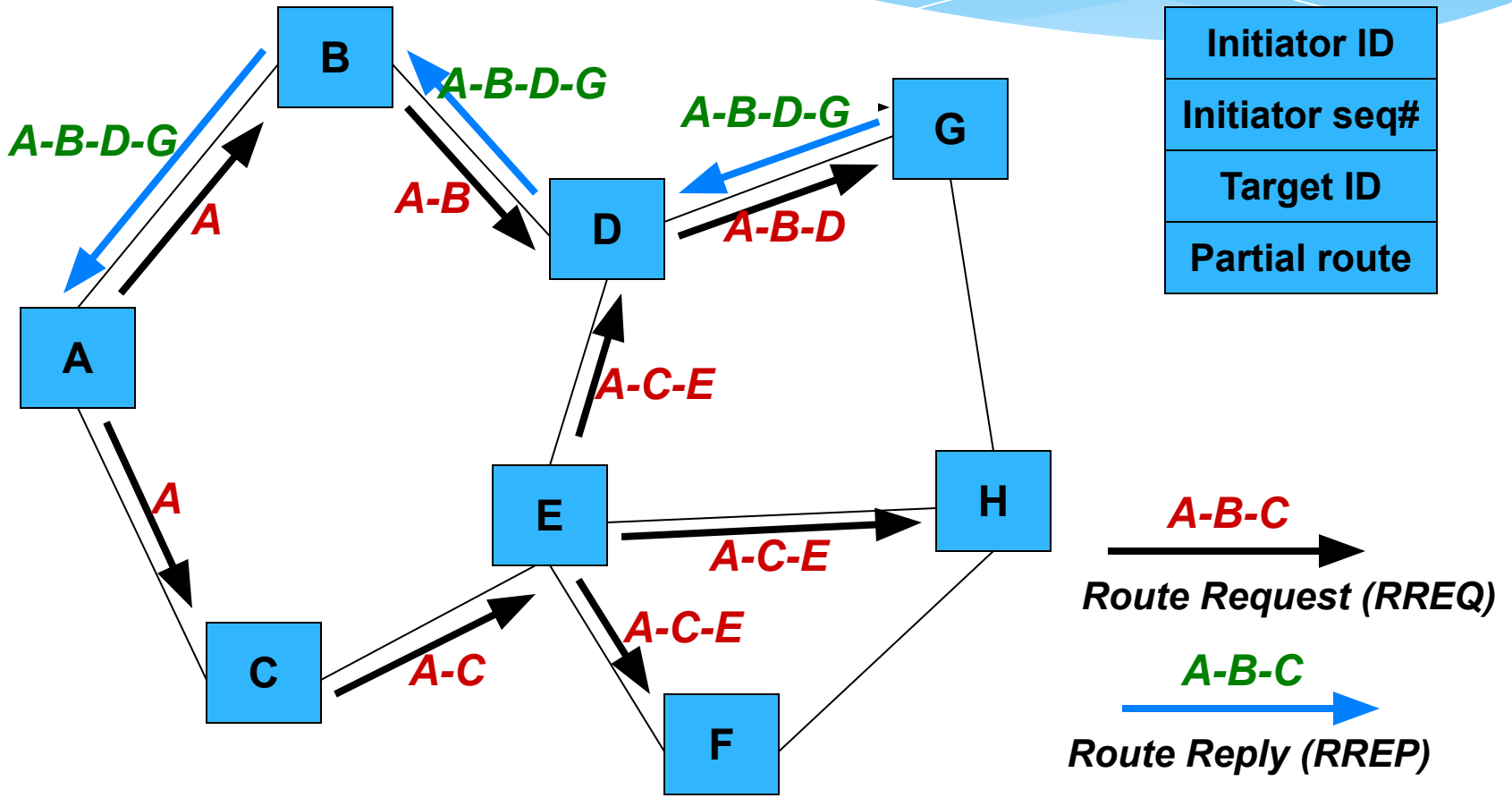
Dynamic Source Routing: Advantages

* Routes maintained only between nodes who need to communicate
  * reduces overhead of route maintenance
* Route caching can further reduce route discovery overhead
* A single route discovery may yield many routes to the destination, due to intermediate nodes replying from local caches

Dynamic Source Routing: Disadvantages

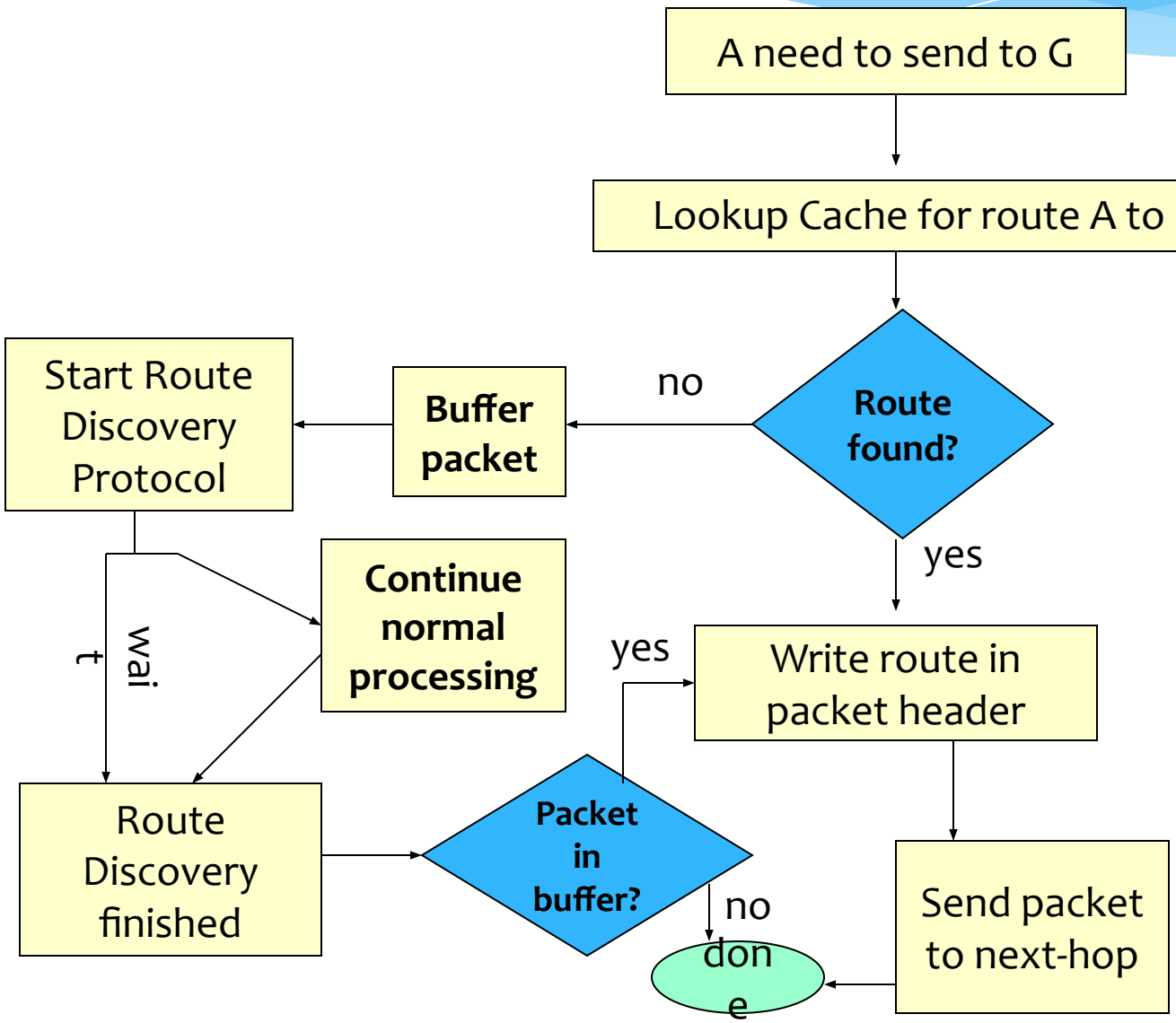
* Packet header size grows with route length due to source routing
* Flood of route requests may potentially reach all nodes in the network
* Potential collisions between route requests propagated by neighboring nodes
  * insertion of random delays before forwarding RREQ
* Increased contention if too many route replies come back due to nodes replying using their local cache
  * Route Reply *Storm* problem
* Stale caches will lead to increased overhead

Route Discovery

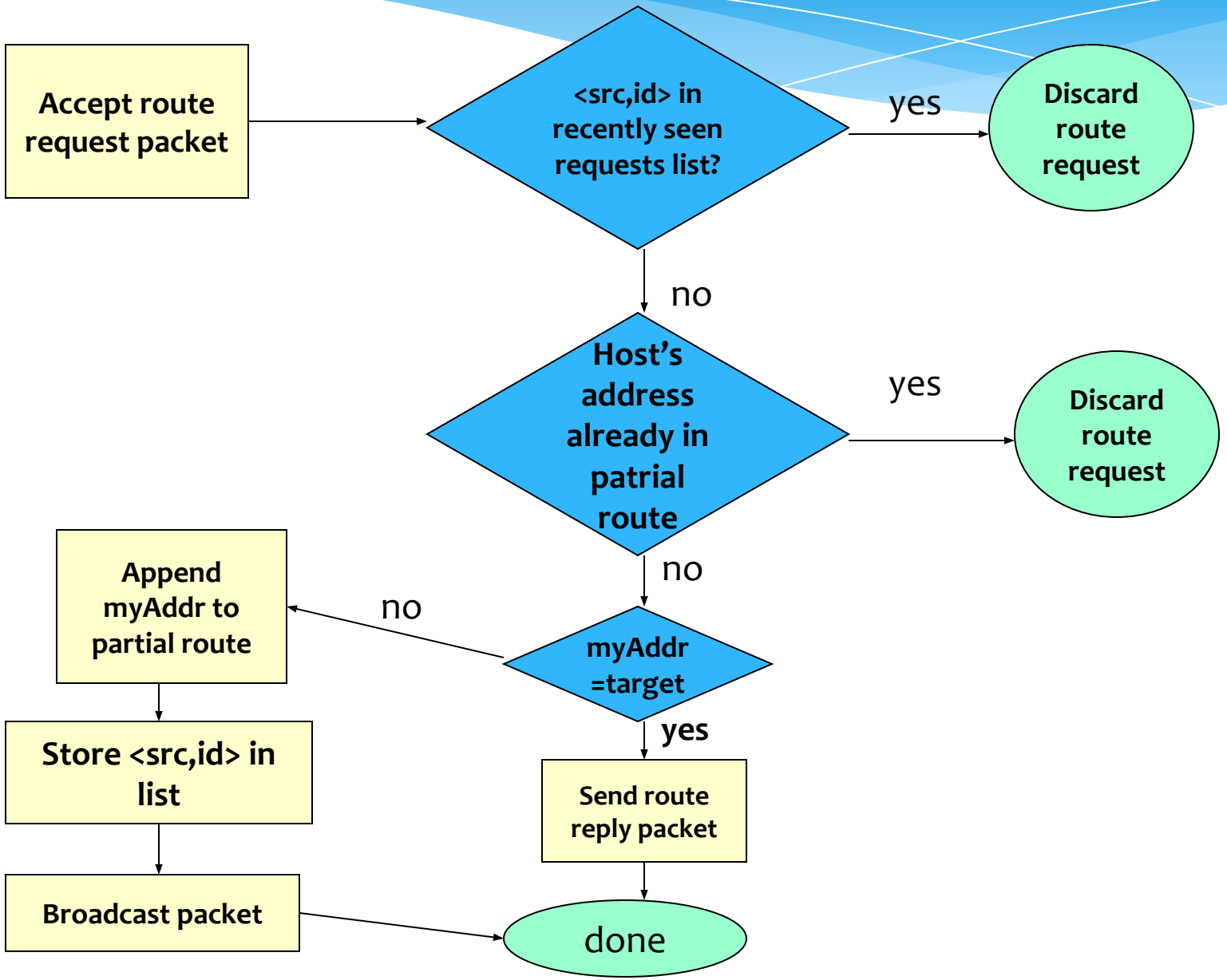


Route Discovery is issued with exponential back-off intervals.

Route Discovery: at source A



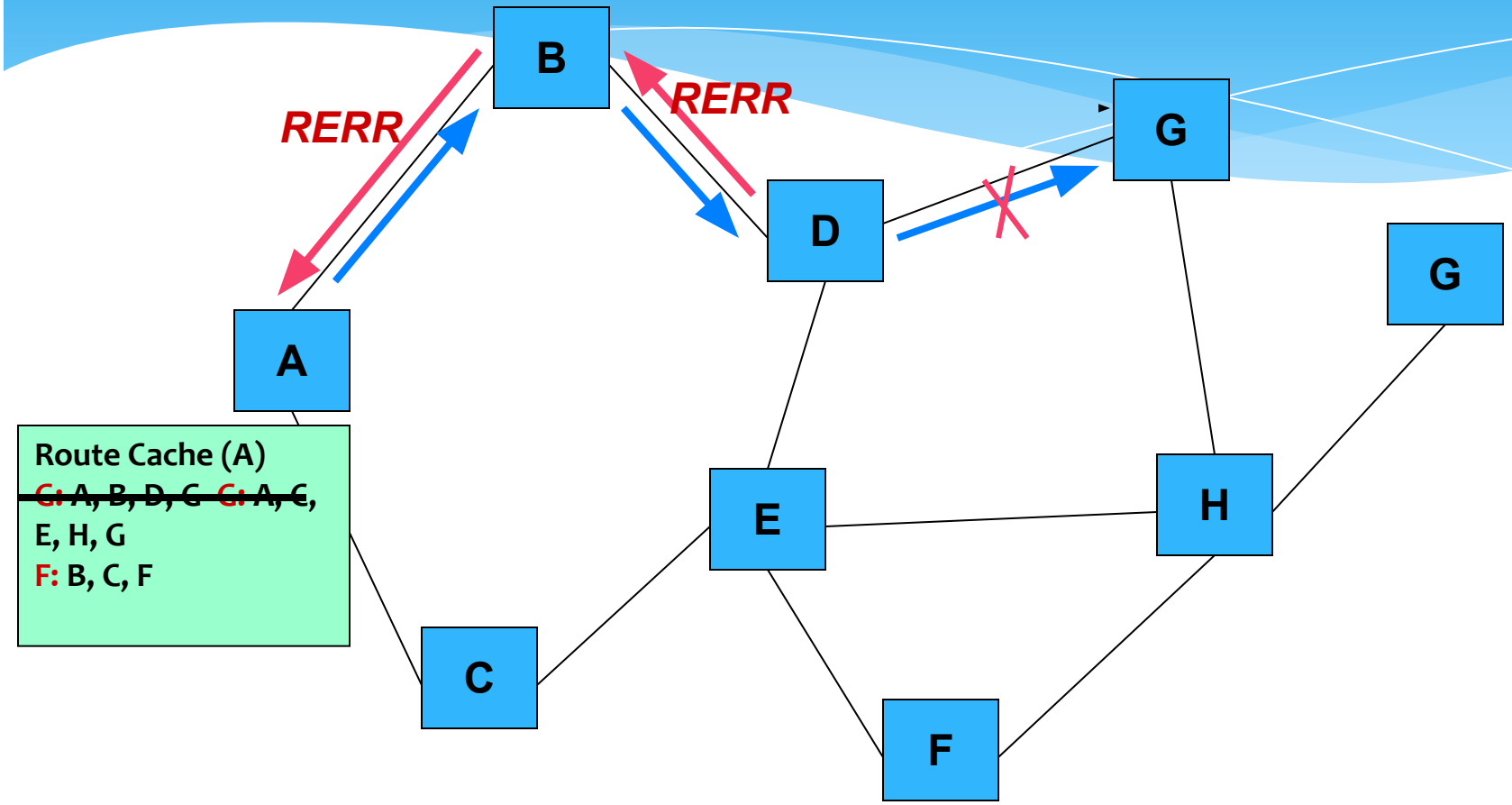
Route Discovery: At an intermediate node



Route Maintenance

* Route maintenance performed only while route is in use
* Error detection:
  * Monitors the validity of existing routes by *passively* listening to data packets transmitted at neighboring nodes
  * Lower level acknowledgements
* When problem detected, send *Route Error* packet to original sender to perform new route discovery
  * Host detects the error and the host it was attempting;
  * *Route Error* is sent back to the sender the packet – original src

Route Maintenance



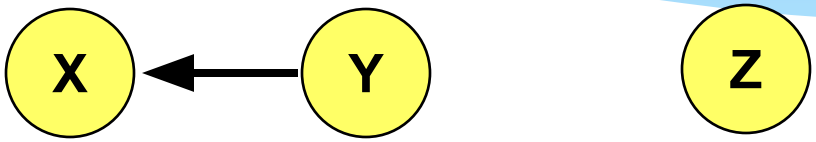
Destination Sequenced Distance Vector (DSDV)

* Based on Bellman-Ford algorithm
* Enhanced with sequence number to avoid loops
  * Fresher routes have higher sequence numbers
* Optimizations added to reduce routing overheads – incremental data exchange, delayed exchange of updates

Destination-Sequenced Distance-Vector (DSDV)

* Each node maintains a routing table that stores
  * next hop, cost metric towards each destination
  * a sequence number that is created by the destination itself
* Each node periodically forwards routing table to neighbors
  * Each node increments and appends its sequence number when sending its local routing table
* Each route is tagged with a sequence number; routes with greater sequence numbers are preferred
* Each node advertises a monotonically increasing even sequence number for itself
* Traffic overhead even if there is no change in network topology
* Maintains routes which are never used

Destination-Sequenced Distance-Vector



* When X receives information from Y about a route to Z
  * Let destination sequence number for Z at X be S(X), S(Y) is sent from Y
  * If S(X) > S(Y), then X ignores the routing information received from Y
  * If S(X) = S(Y), and cost of going through Y is smaller than the route known to X, then X sets Y as the next hop to Z
  * If S(X) < S(Y), then X sets Y as the next hop to Z, and S(X) is updated to equal S(Y)

Protocol Overview

* Updates are transmitted periodically and when there is any significant topology change
  * Make immediate route advertisement on significant changes in routing table
  * but wait with advertising of unstable routes (damping fluctuations)
* In case of broken edge, it increments the sequence number of the route and advertises it with infinite metric

DSDV (Table Entries)

Destination	Next	Metric	Seq. Nr	Install Time	Stable Data
A	A	0	A-550	001000	Ptr_A
B	B	1	B-102	001200	Ptr_B
C	B	3	C-588	001200	Ptr_C
D	B	4	D-312	001200	Ptr_D

* **Sequence number** originated from destination. Ensures loop freeness.
* **Install Time** when entry was made (used to delete stale entries from table)
* **Stable Data** Pointer to a table holding information on how stable a route is. Used to damp fluctuations in network.

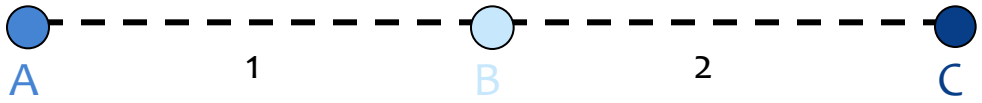
DSDV (Route Advertisements)

* Advertise to each neighbor own routing information
  * Destination Address
  * Metric = Number of Hops to Destination
  * Destination Sequence Number
* Rules to set sequence number information
  * On each advertisement increase own destination sequence number (use only even numbers)
  * If a node is no more reachable (timeout) increase sequence number of this node by 1 (odd sequence number) and set metric = ∞

DSDV (Route Selection)

* Update information is compared to own routing table
  * 1. Select route with higher destination sequence number (This ensure to use always newest information from destination)
  * 2. Select the route with better metric when sequence numbers are equal.

DSDV (Tables)



Dest.	Next	Metric	Seq
A	A	0	A-550
B	B	1	B-100
C	B	3	C-586

Dest.	Next	Metric	Seq
A	A	1	A-550
B	B	0	B-100
C	C	2	C-588

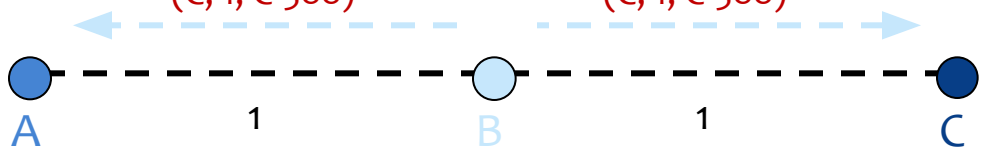
Dest.	Next	Metric	Seq.
A	B	3	A-550
B	B	2	B-100
C	C	0	C-588

DSDV (Route Advertisement)

B increases Seq.Nr from 100 -> 102
B broadcasts routing information
to Neighbors A, C including destination
sequence numbers

(A, 1, A-550)
(B, 0, B-102)
(C, 1, C-588)

(A, 1, A-550)
(B, 0, B-102)
(C, 1, C-588)



Dest.	Next	Metric	Seq
A	A	0	A-550
B	B	1	B-102
C	B	2	C-588

Dest.	Next	Metric	Seq
A	A	1	A-550
B	B	0	B-102
C	C	1	C-588

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-102
C	C	0	C-588

DSDV (Respond to Topology Changes)

* Immediate advertisements
  * Information on new Routes, broken Links, metric change is immediately propagated to neighbors.

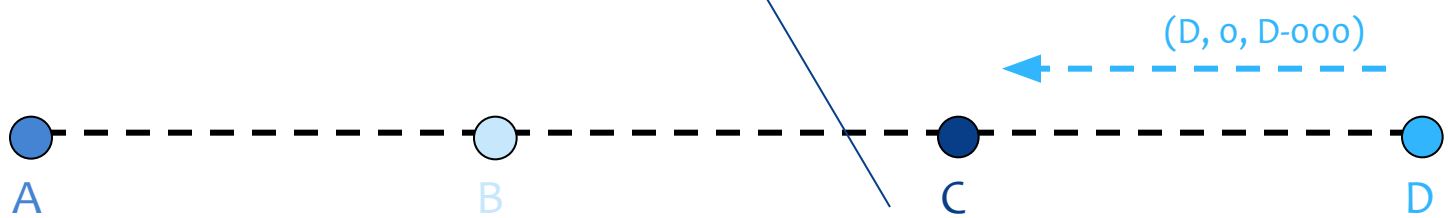
* Full/Incremental Update:
  * Full Update: Send all routing information from own table.

  * Incremental Update: Send only entries that has changed. (Make it fit into one single packet)

DSDV (New Node)

2. Insert entry for D with sequence number D-000
Then immediately broadcast own table

1. D broadcast for first time
Send Sequence number D-000



Dest.	Next	Metric	Seq.
A	A	0	A-550
B	B	1	B-104
C	B	2	C-590

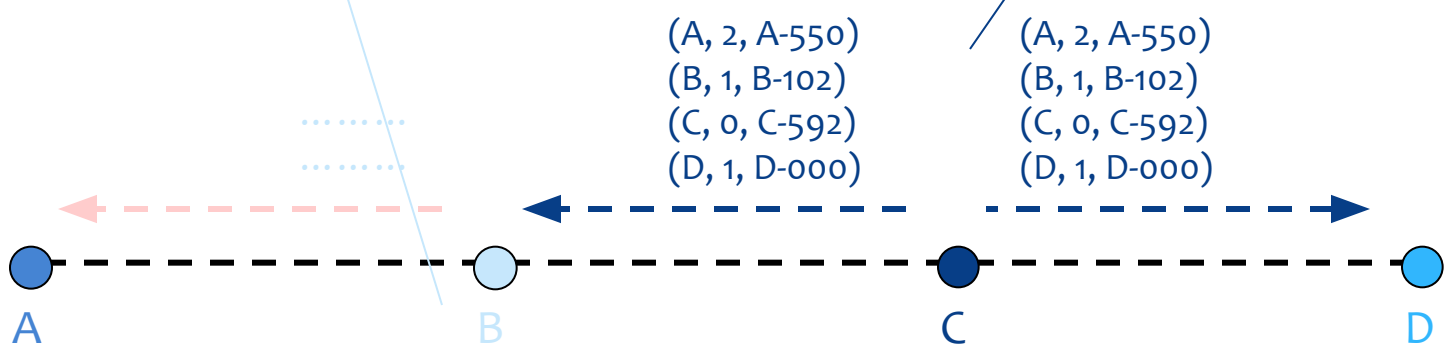
Dest.	Next	Metric	Seq.
A	A	1	A-550
B	B	0	B-104
C	C	1	C-590

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-104
C	C	0	C-590
D	D	1	D-000

DSDV (New Node cont.)

3. C increases its sequence number to C-592 then broadcasts its new table.

4. B gets this new information and updates its table…….



Dest.	Next	Metric	Seq.
A	A	0	A-550
B	B	1	B-104
C	B	2	C-590

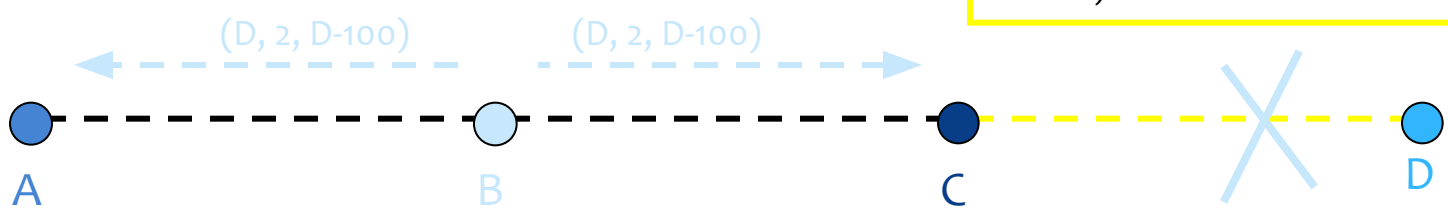
Dest.	Next	Metric	Seq.
A	A	1	A-550
B	B	0	B-102
C	C	1	C-592
D	C	2	D-000

Dest.	Next	Metric	Seq.
A	B	2	A-550
B	B	1	B-102
C	C	0	C-592
D	D	1	D-000

DSDV (no loops, no count to infinity)

2. B does its broadcast
-> no affect on C (C knows that B has stale information because C has higher seq. number for destination D)
-> no loop -> no count to infinity

1. Node C detects broken Link:
-> Increase Seq. Nr. by 1
(only case where not the destination sets the sequence number -> odd number)



Dest.	Next	Metric	Seq.
...	...	...	
D	B	3	D-100

Dest.c	Next	Metric	Seq.
...	...	...	
D	C	2	D-100

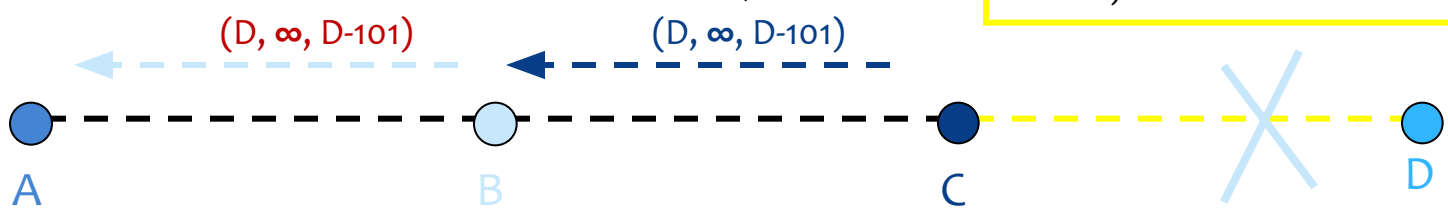
Dest.	Next	Metric	Seq.
...	...	...	
D	D	∞	D-101

DSDV (Immediate Advertisement)

3. Immediate propagation
B to A:
(update information has higher Seq. Nr. -> replace table entry)

2. Immediate propagation
C to B:
(update information has higher Seq. Nr. -> replace table entry)

1. Node C detects broken Link:
-> Increase Seq. Nr. by 1
(only case where not the destination sets the sequence number -> odd number)



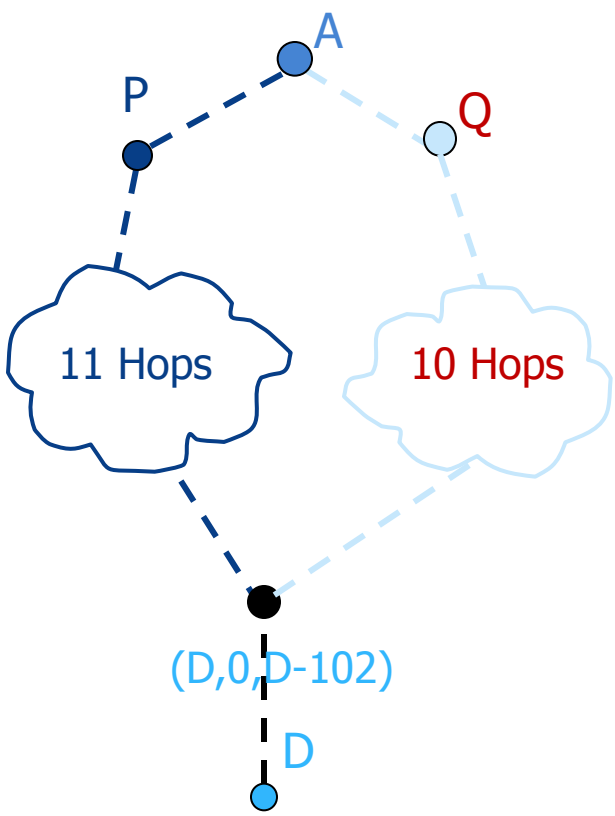
Dest.	Next	Metric	Seq.
...	...	...	...
~~D~~	~~B~~	~~3~~	~~D-100~~
D	B	∞	D-101

Dest.c	Next	Metric	Seq.
...	...	...	...
~~D~~	~~C~~	~~2~~	~~D-100~~
D	C	∞	D-101

Dest.	Next	Metric	Seq.
...	...	...	
~~D~~	~~D~~	~~1~~	~~D-100~~
D	D	∞	D-101

DSDV (Problem of Fluctuations)

What are Fluctuations



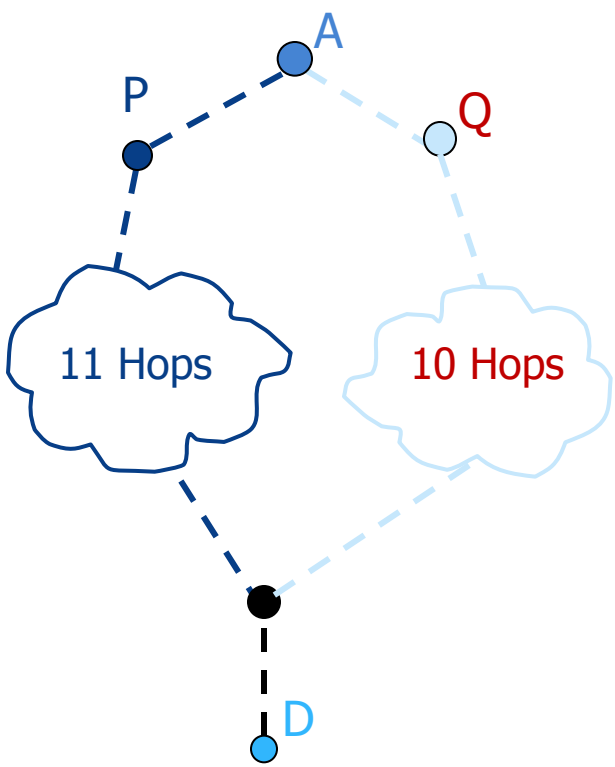
* Entry for D in A: [D, Q, 14, D-100]

* D makes Broadcast with Seq. Nr. D-102
* A receives from P Update (D, 15, D-102)
  -> Entry for D in A: [D, P, 15, D-102]
  A must propagate this route immediately.
* A receives from Q Update (D, 14, D-102)
  -> Entry for D in A: [D, Q, 14, D-102]
  A must propagate this route immediately.

This can happen every time D or any other node does its broadcast and lead to unnecessary route advertisements in the network, so called fluctuations.

DSDV (Damping Fluctuations)

How to damp fluctuations



* Record last and avg. Settling Time of every Route in a separate table. (Stable Data) Settling Time = Time between arrival of first route and the best route with a given seq. nr.

* A still must update his routing table on the first arrival of a route with a newer seq. nr., but he can wait to advertising it. Time to wait is proposed to be 2*(avg. Settling Time).

* Like this fluctuations in larger networks can be damped to avoid unnecessary advertisement, thus saving bandwidth.

Summary

* Advantages
  * Simple (almost like Distance Vector)
  * Loop free through destination seq. numbers
  * No latency caused by route discovery

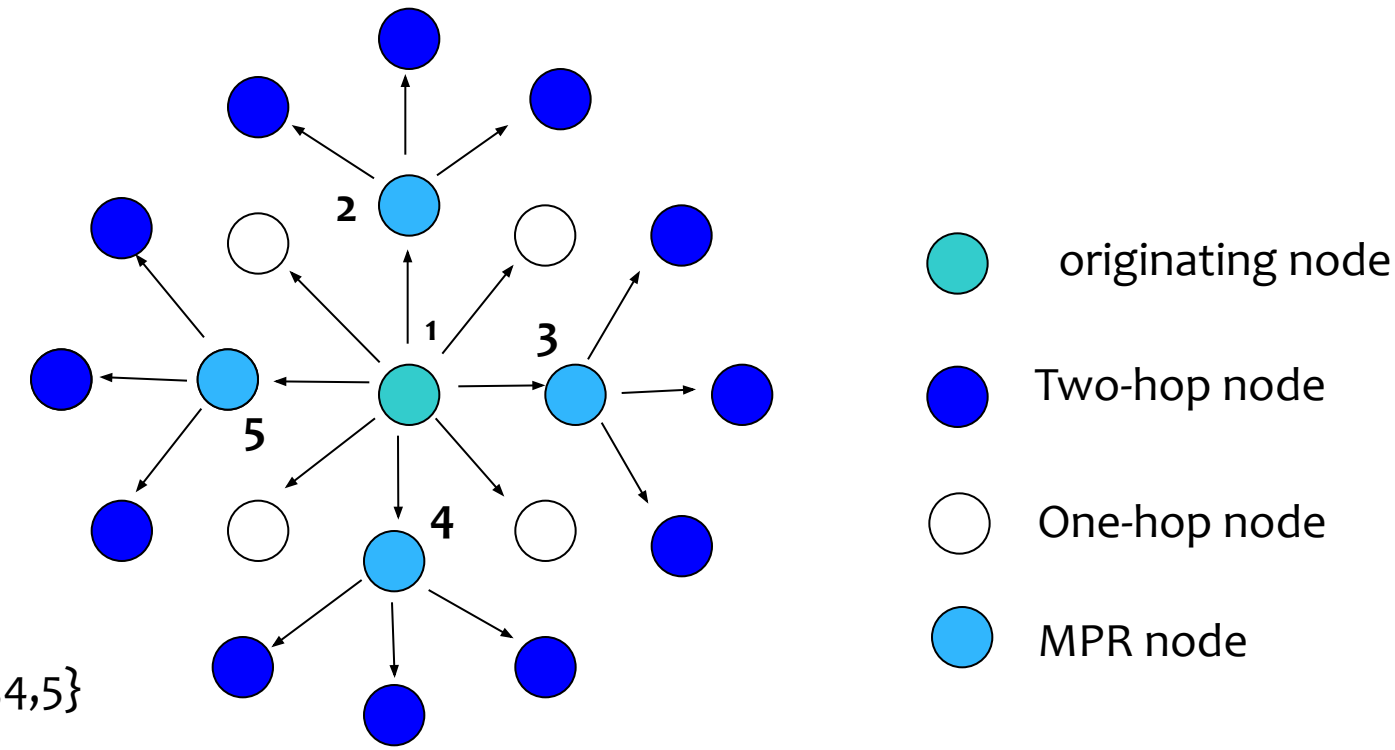
* Disadvantages
  * No sleeping nodes
  * Overhead: most routing information never used

Optimized Link State Routing

* Proactive, Table driven, based on Link State approach
* Motivation: To reduce traffic through selective Flooding
* Process:
  * Reduces the size of the control pkts; as it includes a subset of links with its neighbours who are it's multi-point relay (MPR) selectors.
  * Only MPRs are participating in flooding [Selective Flooding]
  * Multipoint relays of A are its neighbors such that each two-hop neighbor of A is a one-hop neighbor of one multipoint relay of A. The multi point relay set of node N, such that each node in two hop neighborhood of N must have bidirectional links towards the members of MPR (N).
  * MPRs are selected in a way such that it covers all 2-hop away nodes. Reduces duplicate transmission within same range.

Optimized link source routing

Type: Proactive; suitable for large & dense network for better optimization compared to traditional Link-State



MPR(1)={2,3,4,5}

Optimized Link State Routing

* Generates normal periodic messages; no extra messages for route error etc
* Keeps routes for all destinations; performs better when significant data exchange takes place and the (source, destination) pair changes with time.
* Suitable for large and dense networks; more optimization is achieved with greater density
* Does not require a reliable transmission; does not need an in-order delivery as the control pkts are time-stamped with seq no

OLSR

* Mobility ?
* In case of nodes mobility, packet can be successfully delivered if its speed is such that it’s movement could be followed by the neighbourhood, at least. It supports mobility iff the mobility can be traced by the local control message i.e. The frequency of the message.

OLSR: Neighbour Sensing

* Each node finds neighbour nodes with which it has a direct & bidirectional link
* Hello messages broadcast periodically
  * List of address of neighbours having bi-directional link
  * List of address of the neighbours which are heard by this node

  * Hello msg permits each node to learn the knowledge up to 2 –hop neighbours. Each node selects MPRs. Includes the information in Hello message.
  * Creates a neighbour table : 1 hop neighbours, Status, list of 2-hop neighbours reachable by this 1 –hop neighbour, holding time (validity), Sequence no for identifying the most recent MPR set

OLSR: Topology Control

* Multipoint Relay Sets are recalculated
  * A change in the neighbourhood is detected when a bidirectional link with a neighbour is failed/ a new link appears with bi-directional status/ Change in the 2-hop neighbour set with bi-directional link
  * TC message declared by each node that contains the MPR set of that node
  * Topology table (Add of a potential destination i.e. an MPR selector in the received TC msg, Add of last hop i.e. Originator of TC msg. Sequence No, Holding time)

Reconfigurable Wireless Network (RWN)

* An Adhoc Network that can be rapidly deployed without relying on a pre-existing fixed network.
* The nodes can join/leave frequently without any disruption of other node's communication.
* Nodes are highly mobile; rapidly change the node constellation and the presence/ absense of links
* Large networks (hundreds to thousand nodes); Significant variations in nodal speed, direction of movement; restrictions on paths

RWN

* RWN s are distinguished from other ad-hoc by rapidly changing topologies, influenced by network size & node mobility.
* Typical applications like Rescue Mission; Tactical operation, Education
* In spite of such volatility, RWN is expected to deliver diverse traffic types ranging from pure voice to integrated voice, image and even possibly some limited video

RWN: Communication Model

* Nodes are equipped with communication devices; powered by limited batteries.....results in restrict transmission range, communication activity etc.
* Connectivity between nodes is not a transitive relation
* A hierarchy may improve routing/ mobility management in terms of latency but vulnerable to topological reconfigurations.
* Nodes have fixed Ids and have equal capabilities.
* It should allow communication between any two nodes; but it is envisioned that a large traffic will be between the nodes those are geographically closed.

RWN: Routing Features

- Robust routing & Mobility Management to increase the reliability and availability
- Adaptive algorithms to adjust frequently changing radio propagation, network and traffic conditions
- Low overhead algorithms to preserve radio communication resource
- Multiple routes maintenance between a pair of nodes to avoid congestion.....increases reliability and survivability

RWN: Routing Features

- RWN.....large geographical network size, large no. of nodes, fast & varying nodal movement, frequent topological changes
- Require multiple, loop free routes
- Flat routing
- Optimal routes to be discovered
- Reduces the threat of network congestion

Traditional Routing Schemes

- As seen, the amount of update traffic is quite high in proactive routing.
- In reactive routing, route query is delayed because route information may not be available

  ZRP aims to address the problems by combining the best properties of both approaches.

-

ZRP: Motivation

- ZRP limits the scope of Proactive routing to the nodes local neighborhood.
  - Creation of a new link at one end of the network is an important local event; may not be a significant for all the nodes. Thus Instead of informing all, inform only the local neighborhoods; cost will be less.
- Search can be global and that could be done by querying only a subset of nodes.

The Zone Routing Protocol

Based on the concept of zones

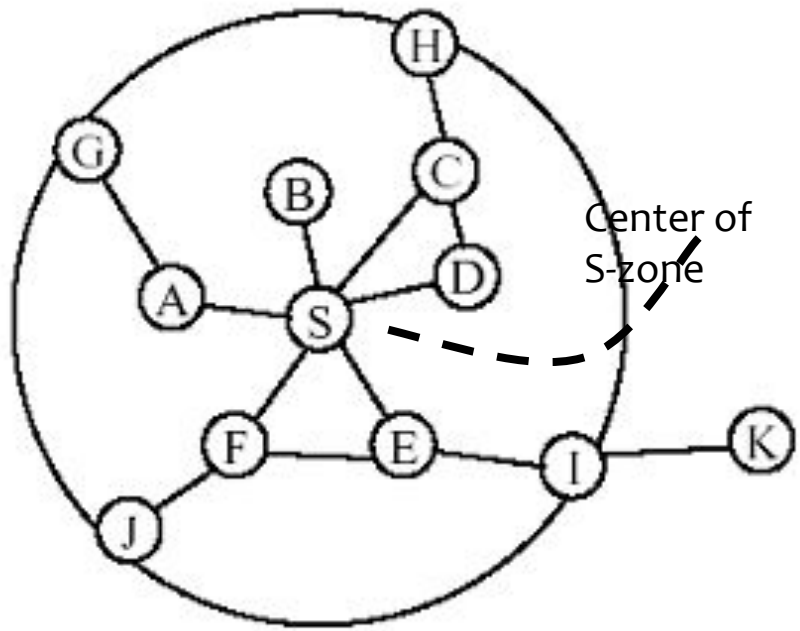
- A zone is defined (how?) for each node separately
- Zone radius r given as n
- The zones overlap

- Peripheral nodes

  G, H, J, and I

- Neighbor nodes

  B, C, ….



r = 2 (hops)

We depict zones as circles, but they are not.

The Zone Routing Protocol

Based on the concept of zones

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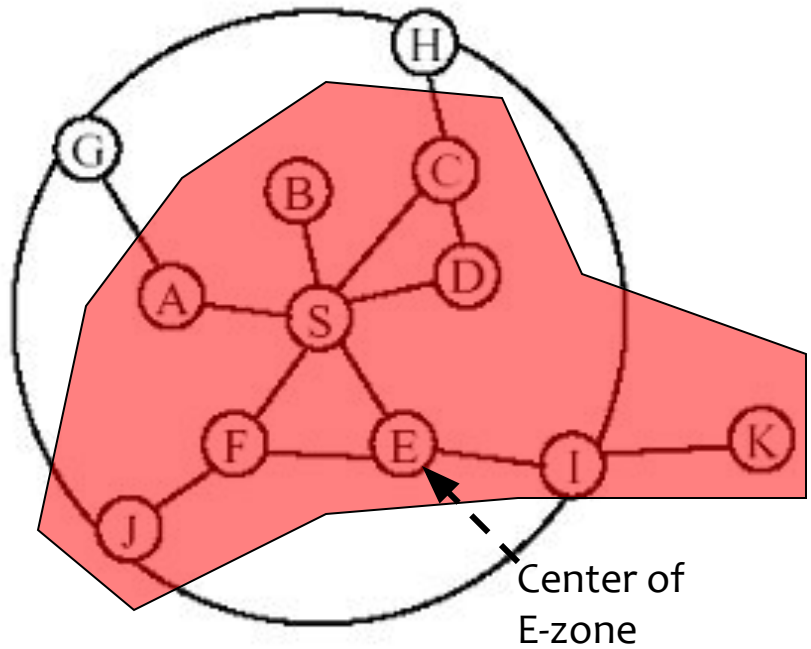


r = 2 (hops)

We depict zones as circles, but they are not.

Proactive Intra-zone Routing

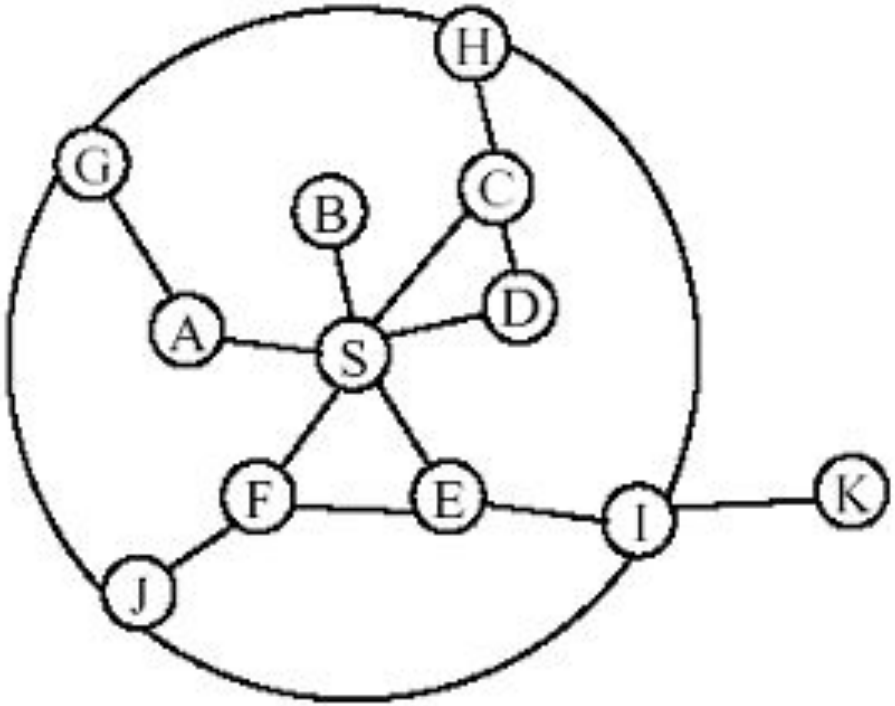
ZRP refers to the locally proactive routing component as **IARP**

IARP is not a specific routing protocols. Instead, IARP is a family of proactive routing protocols.

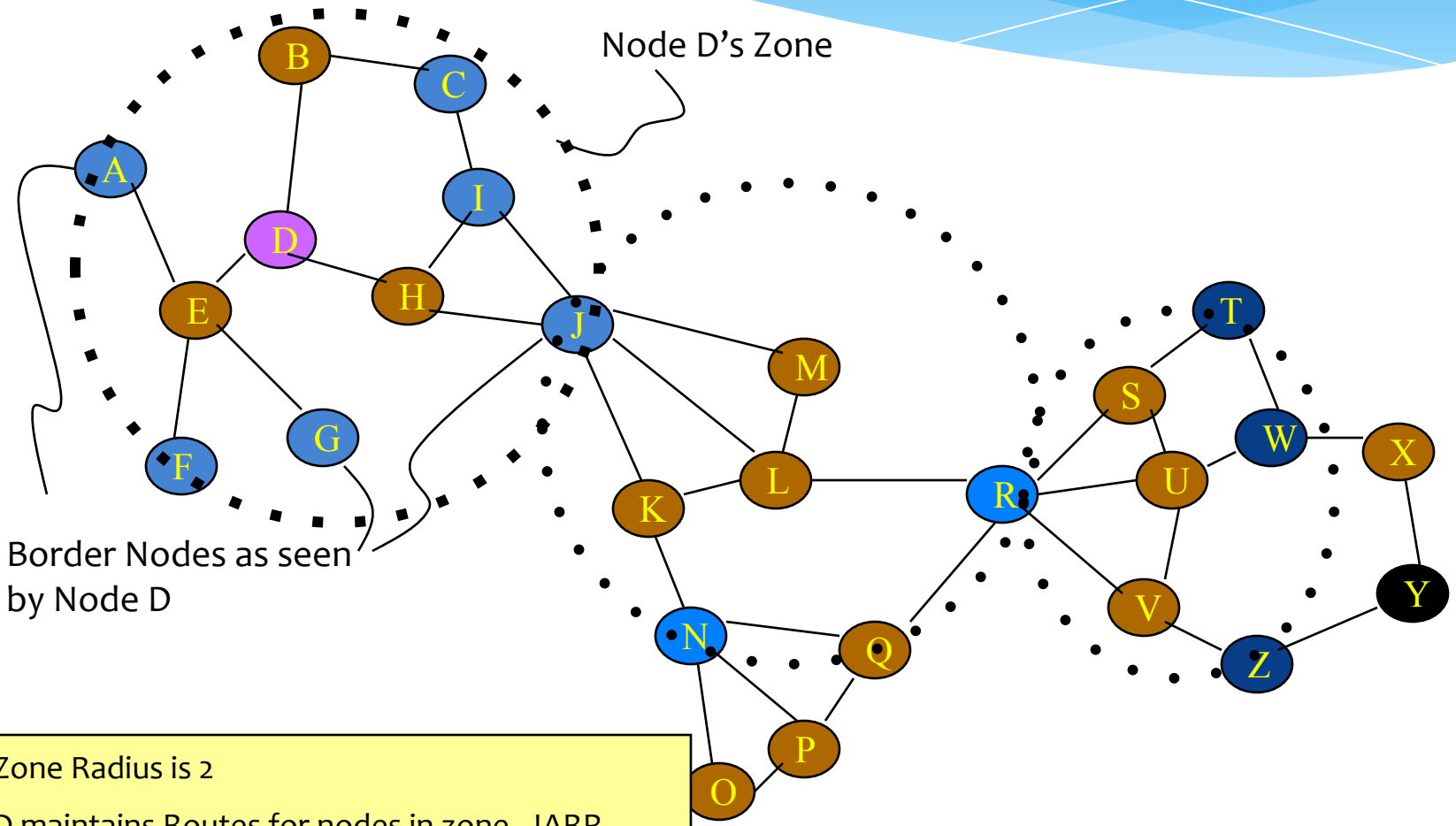
Each node maintains the routing info to all nodes in its zone

The updates are only local in the zone.

Reduces the maintenance costs to a limited zone



How does Intra Zone work ?



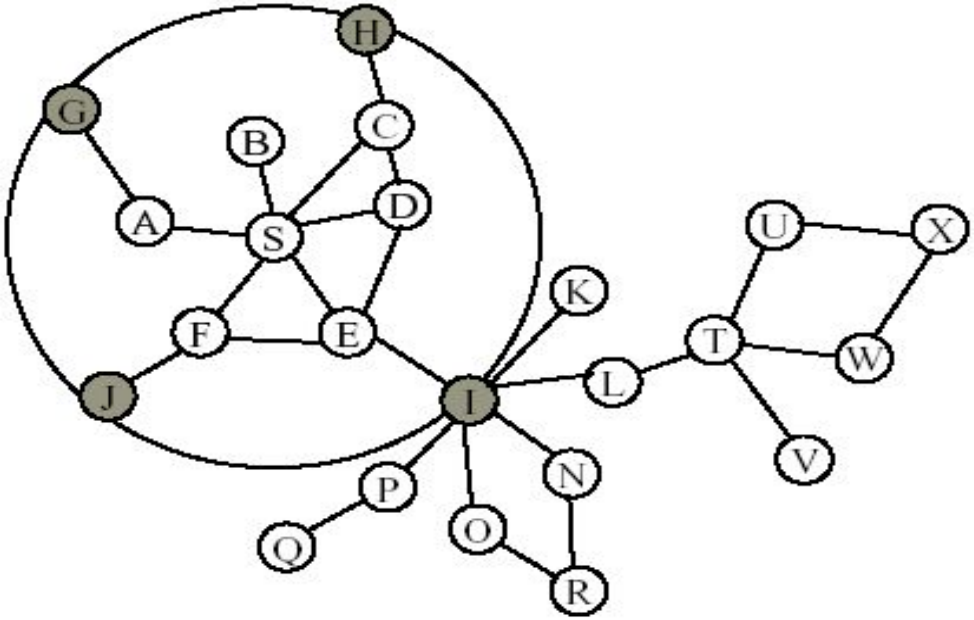
- Zone Radius is 2
- D maintains Routes for nodes in zone --IARP e.g. D knows route to G.
- If node not found, resort to Inter zone search.

Reactive Inter-zone Routing

The globally reactive routing component is named **IERP. IERP** is responsible for **discovering routes to destinations** beyond a node's routing zone.

For example, when S has a packet to send to x, IERP is responsible for discovering this route.

How does IERP work?



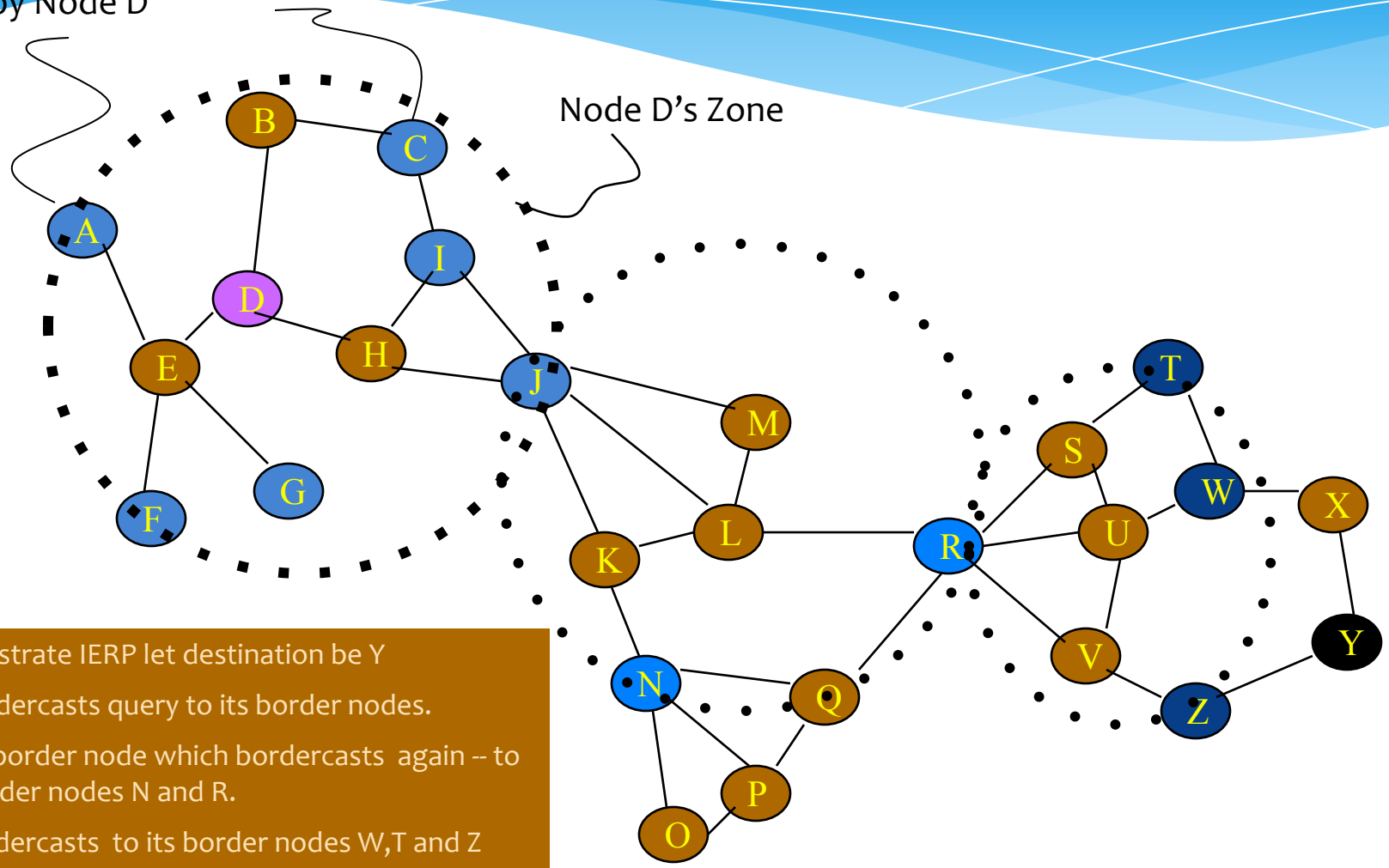
Bordercasting

- Knowledge of local routing zone topology can be exploited to relay a query through the network. [Multicasting]
- The node would direct the query message out only to its peripheral nodes. [based on routing table]
- Compute a border-cast (multicast) tree and append the corresponding packet forwarding Instructions to the bordercast packet.

Bordercasting

- Bordercast nodes would execute the same algorithm that the primary node executed which is:
  - Check to see if the destination can be found within its zone. (How ?).
  - If yes, send a route-reply back to the source, indicating the route to the destination.
  - If not, forward the route-request to its peripheral nodes which executes the same procedure.

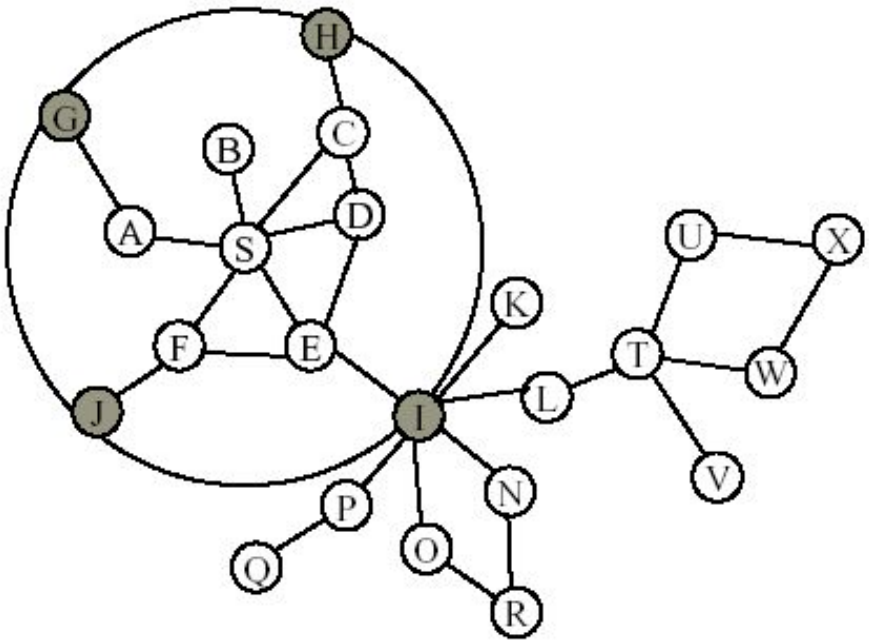
How does IERP work ?



- To illustrate IERP let destination be Y
- D bordercasts query to its border nodes.
- J is a border node which bordercasts again -- to its border nodes N and R.
- R bordercasts to its border nodes W,T and Z
- Y is found.

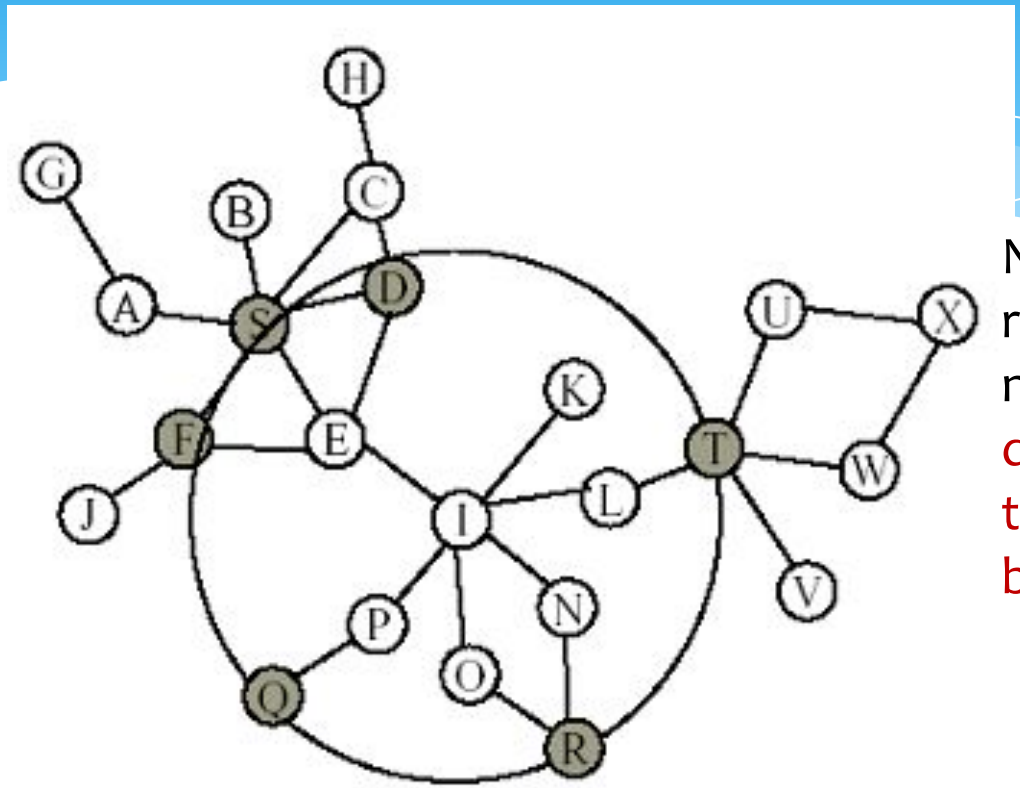
A Concrete Example of How IERP works.

S has a packet to send to x



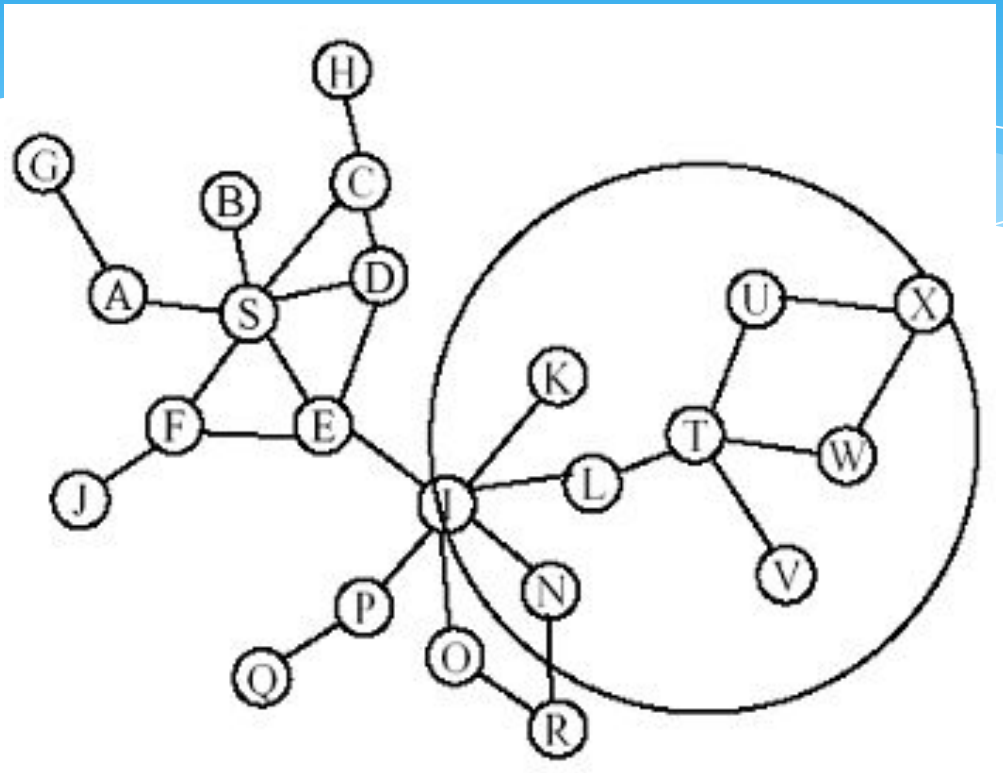
X is not in S's routing zone. S issues a route request using IERP. The request is bordercasted to the peripheral nodes (the gray ones). These gray nodes search their routing table for the destination. They don't see the destination. So the border-casting process continues (at node I, J, G , and H).

An Example (cont.)



Node I bordercasts the request to its peripheral nodes (gray ones). Due to query control mechanisms, the request is not passed back to nodes D, F, and S.

An Example (cont.)



Finally, the route request is received by node T, which can find the destination in its routing zone. A rout reply is sent back to node S.

Discussion: ZRP

Single route query can return multiple route replies; the best route will be selected based on selected metric.

Given sufficient storage, nodes may remove the routing information accumulated in route request packet and store it in a temp route query cache. It reduces the length of query packet, decreases the query response time.

Initiate a global route search only when there is a substantial change for reducing the cost of route discovery. A broken link within an active path may be the cause. Local Path repair procedure is initiated. It replaces a broken link with a minipath between the link ends. A path update is generated and sent to the path's endpoints. It may not be the shortest one. After some repairs, the path endpoints initiate a new route discovery for optimal path.

Discussion: ZRP

Neighboring routing zones heavily overlap, each node may forward a route request multiple times, more control traffic than flooding. The challenge is to steer the query outward from the original routing zone. It depends on routing zone architecture.

Redundant querying occurs when a route request arrives in a previously queried routing zone. Thus the nodes will be capable to detect their early visited zone.

A node will not relay a query packet to a target bordercast recipient either if that recipient lies inside the routing zone of a previously bordercast node or if this node has already relayed the query to this recipient. (Early Termination)

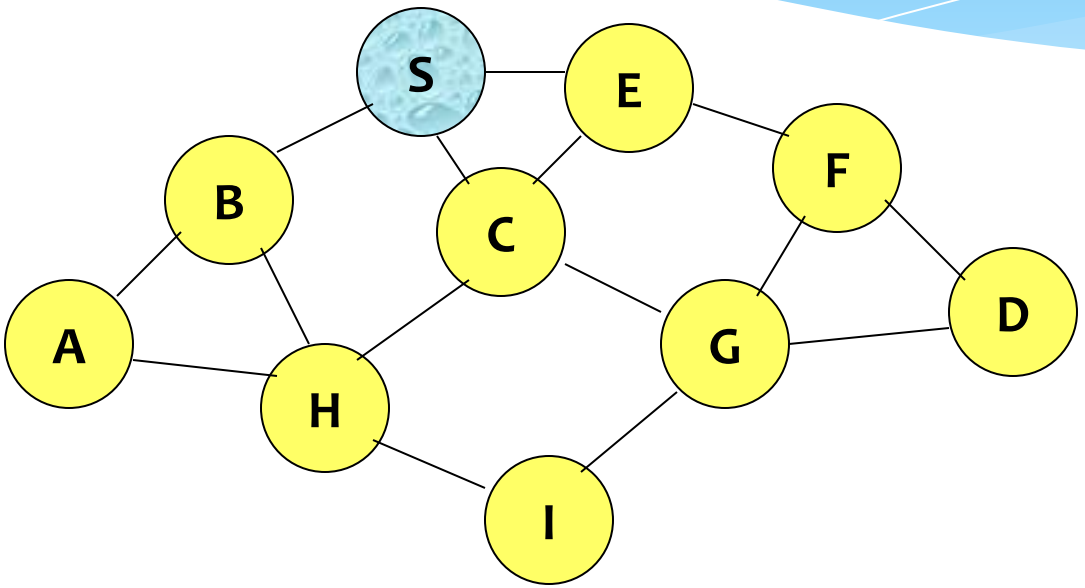
AODV: Motivation

* Ad-hoc On demand Distance Vector Routing
* Objective: Minimalist features: Minimal Control Overhead; Minimal route acquisition latency
* Goal is to reduce the system wise broadcasts (as much as possible) compared to DSDV
* No broadcast ; if the link status does not affect the ongoing communication.
* In DSDV, local movements have global effects. Triggered broadcast is replaced by careful bookkeeping.

AODV

* Route Requests (RREQ) are forwarded in a manner similar to DSR
* It does not add any overhead to the packets carrying data.
* When a node re-broadcasts a Route Request [**source IP, Broadcast ID,** current Seq no, Destination node IP, last known sequence no] & sets up a reverse path pointing towards the source
  * AODV assumes symmetric (bi-directional) links
* When the intended destination receives a Route Request, it replies by sending a Route Reply (RREP)
* Route Reply travels along the reverse path set-up when Route Request is forwarded

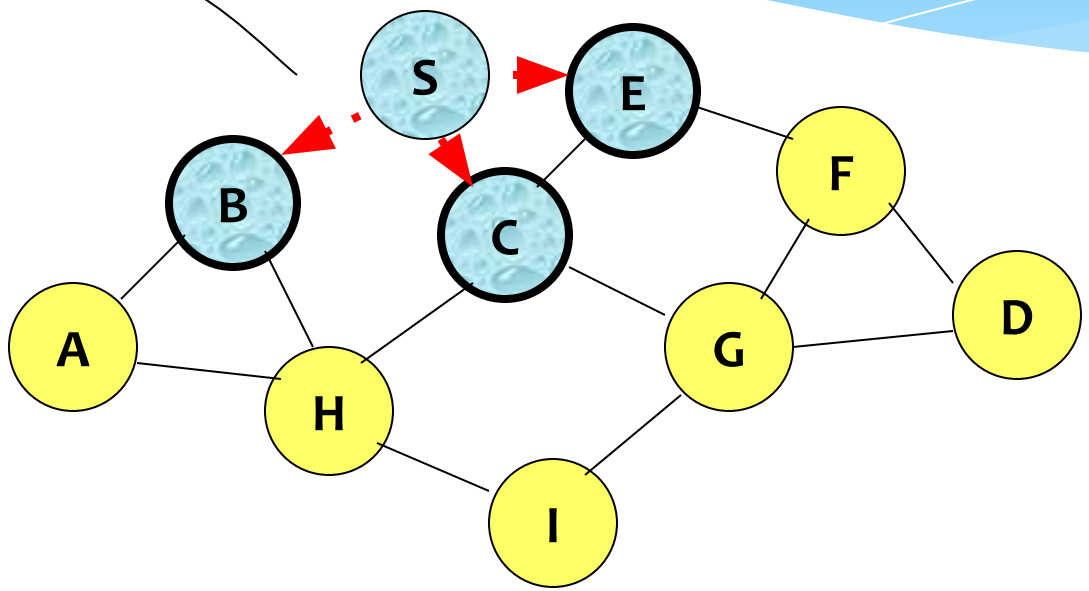
Route Requests in AODV



Represents a node that has received RREQ for D from S

Route Requests in AODV

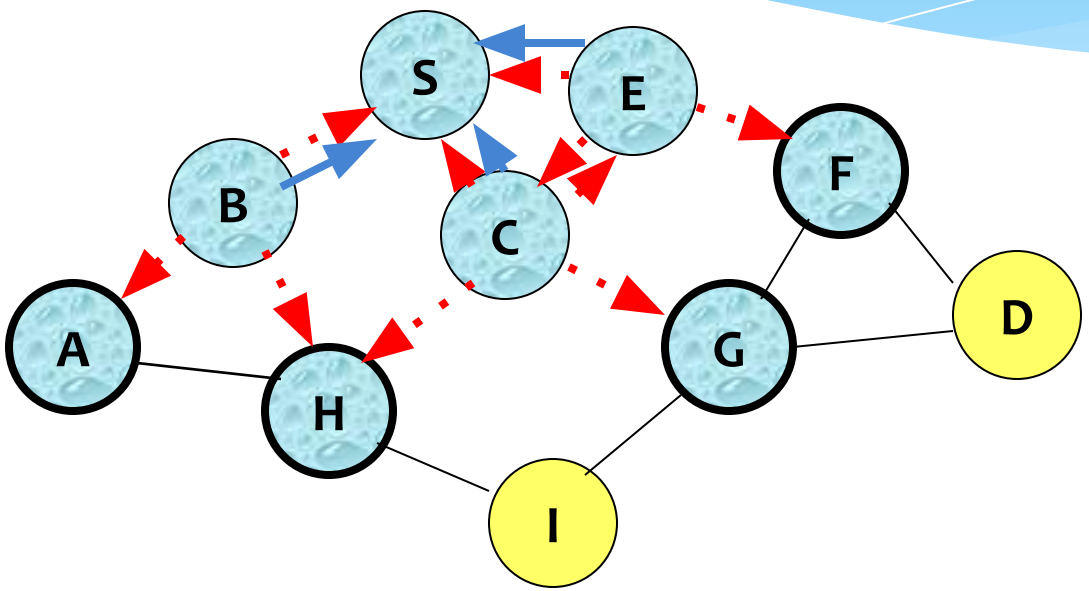
Broadcast transmission



Represents transmission of RREQ

A node re-broadcasts a Route Request [**source IP, Broadcast ID,** current Seq no, Destination node IP, last known sequence no]

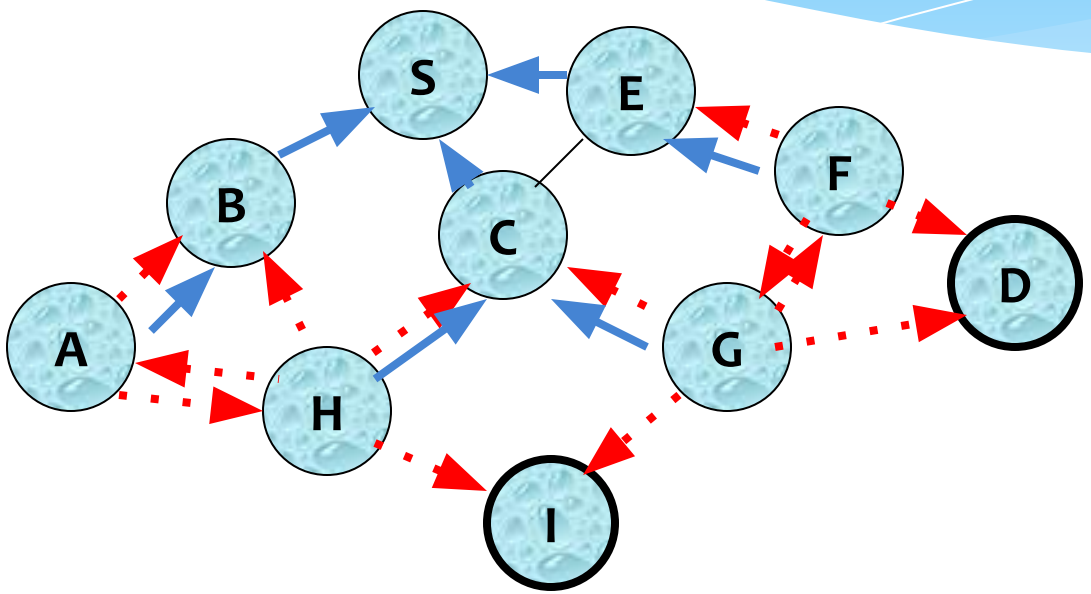
Route Requests in AODV



Represents links on Reverse Path

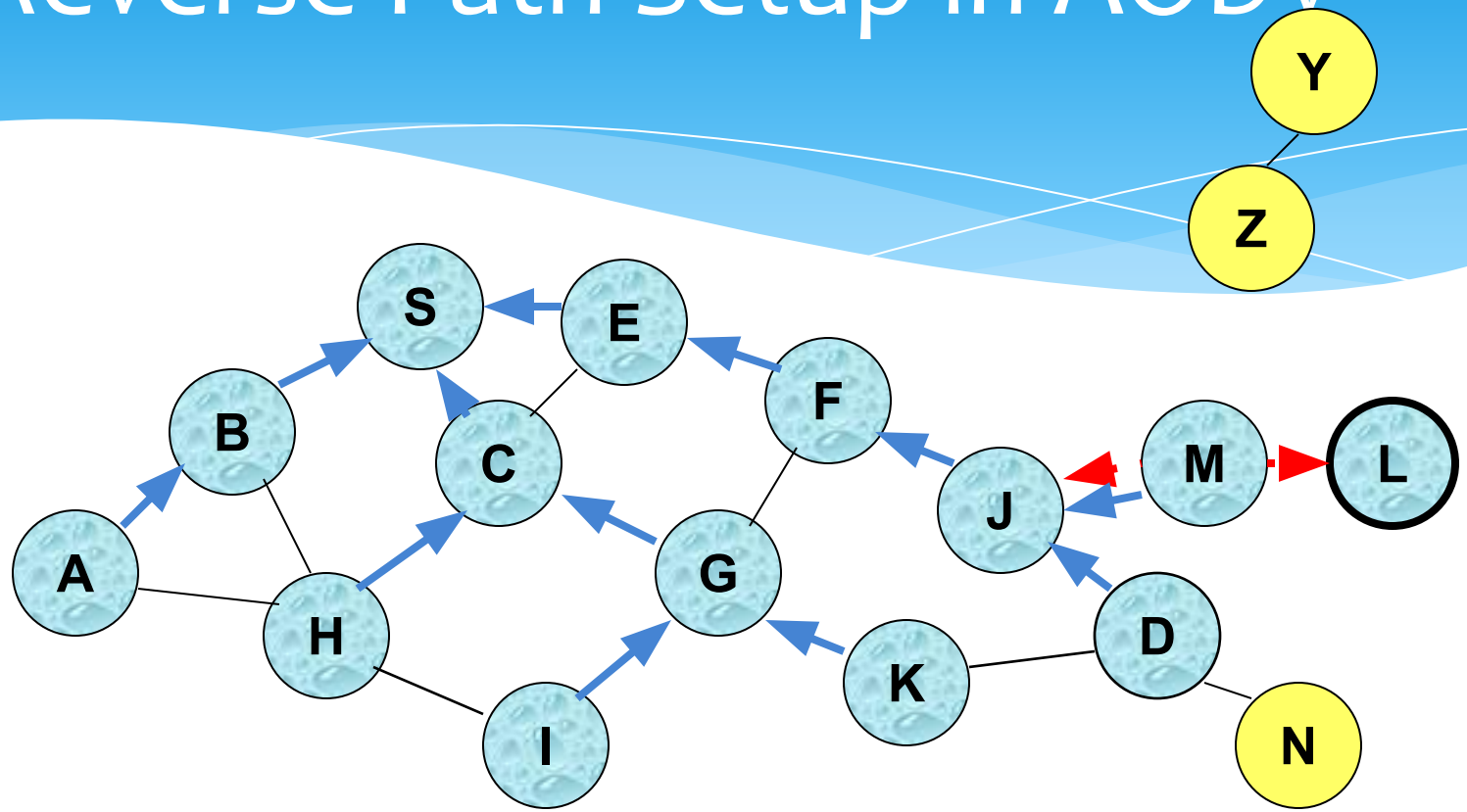
Before re-broadcasting, [source IP, Broadcast ID] will be checked for avoid Duplication; otherwise process it

Reverse Path Setup in AODV



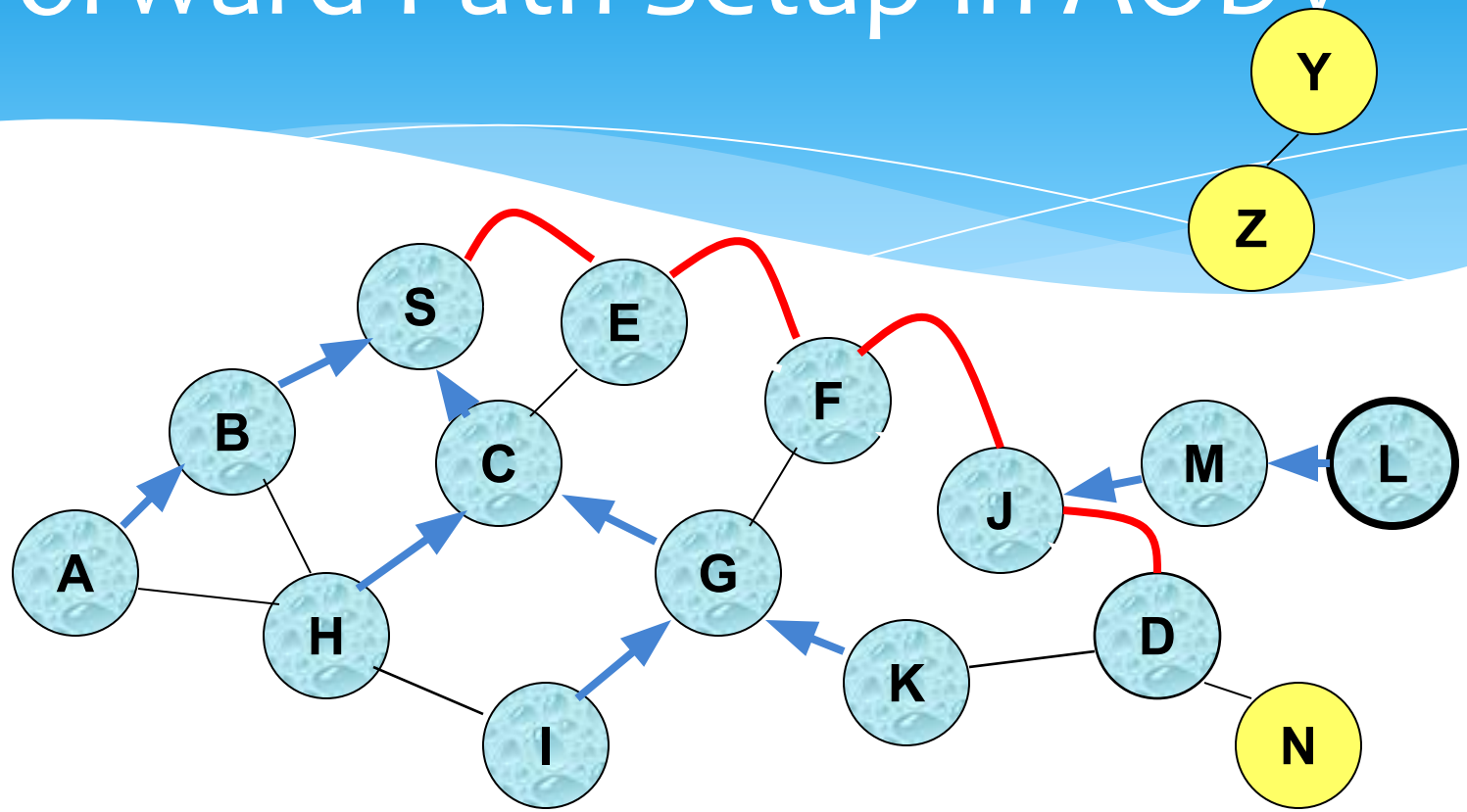
- To process the RREQ, the node sets up a reverse route entry for the source node in the route table. Reverse route entry must have a lifetime For processing the route request; the node have an un-expired entry & an entry within routing table for that dest with equal or greater seq no

Reverse Path Setup in AODV



- Node D **does not forward** RREQ, because node D is the **intended target** of the RREQ

Forward Path Setup in AODV



Forward links are setup when RREP travels along the reverse path

Represents a link on the forward path

AODV – Optimization

* Useful optimization: An intermediate node with a route to D can reply to route request.
  * **Faster operation.**
  * **Multiple routes to be used after discovering .**
  * **Expanding ring search technique.**
  * **Unused routes are expired (through a concept of Aging) and consequently discarded …… route maintenance will be easier.**

* Above optimization can cause loops in presence of link failures

Link Failure

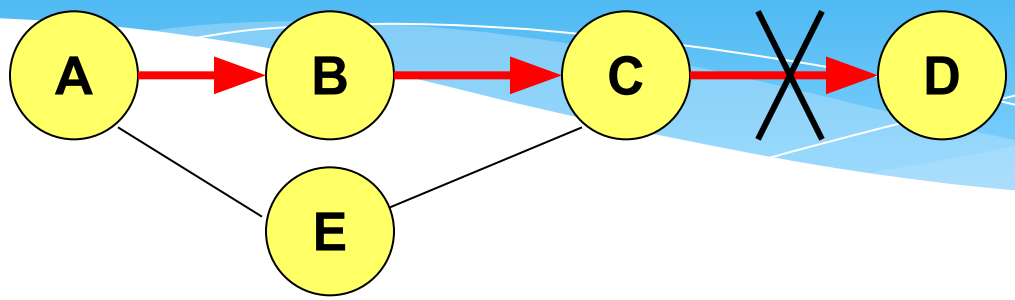
* A neighbor of node X is considered active for a routing table entry if the neighbor sent a packet within *active_route_timeout* interval which was forwarded using that entry

* Neighboring nodes periodically exchange hello message

* When the next hop link in a routing table entry breaks, all active neighbors are informed

* Link failures are propagated by means of Route Error (RERR) messages, which also update destination sequence numbers

AODV: Routing Loops

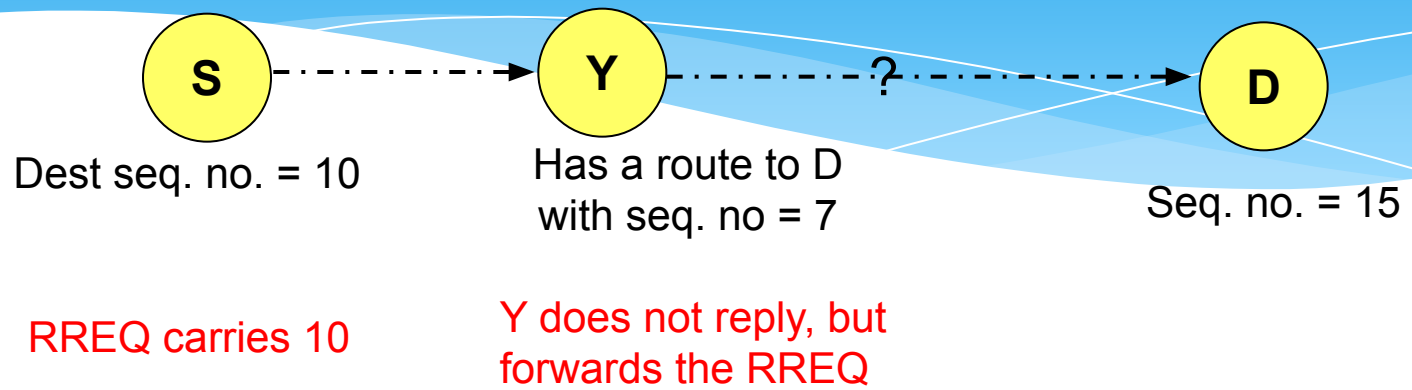


* Assume, link C-D fails, and node A does not know about it (route error packet from C is lost).
* C performs a route discovery for D.
* Node A receives the route request (via path C-E-A)
* Node A replies, since A knows a route to D via node B
* Results in a loop: C-E-A-B-C

AODV: Use Sequence Numbers

* Each node X maintains a sequence number
  * acts as a time stamp
  * incremented every time X sends any message
* Each route to X (at any node Y) also has X's sequence number associated with it, which is Y's latest knowledge of X's sequence number.
* Sequence number signifies 'freshness' of the route – higher the number, more up to date is the route.

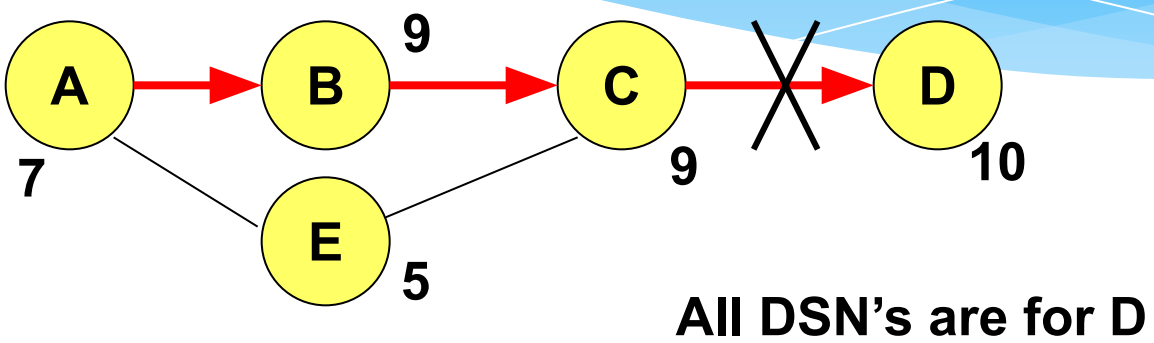
Use of Sequence Numbers in AODV



* Loop freedom: Intermediate node replies with a route (instead of forwarding request) only if it has a route with a higher associated sequence number.

Avoidance of Loop

DSN = Destination Sequence Number.

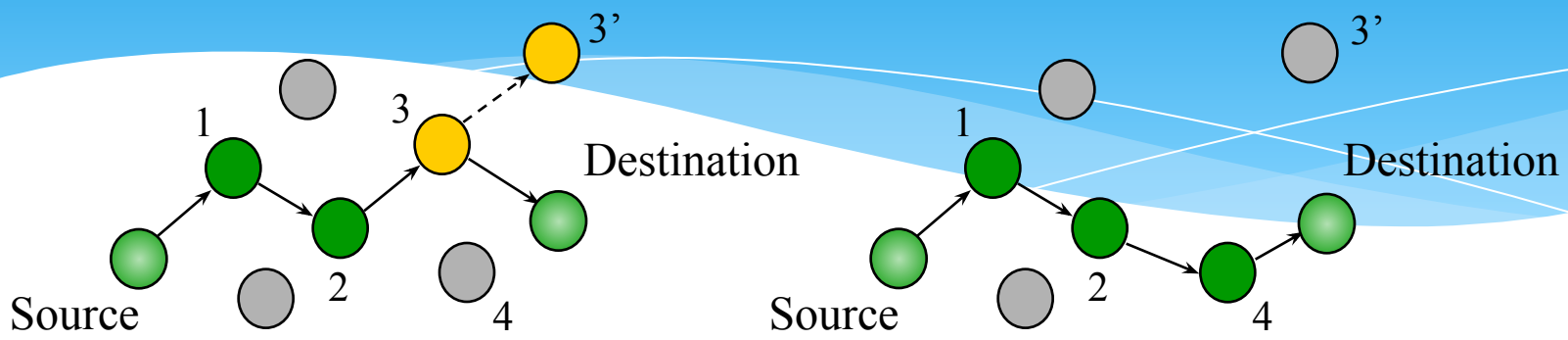


* Link failure increments the DSN at C (now is 10).
* If C needs route to D, RREQ carries the DSN (10).
* A does not reply as its own DSN is less than 10.

Route Maintenance

* Movement of nodes within the ad-hoc network affects only the routes containing those nodes; such a path is called Active Path.
* Route maintenance will be triggered only for active session.
* Route entries with an infinity metric are not immediately deleted as they contain useful current routing information.

Path Maintenance



- Movement not along active path triggers no action
  - If source moves, reinitiate route discovery
- When destination or intermediate node moves
  - upstream node of break broadcasts *Route Error* (RERR)
  - RERR contains list of all destinations no longer reachable due to link break
  - RERR propagated until node with no precursors for destination is reached

AODV: Summary

* Routes need not be included in packet headers

* Nodes maintain routing tables containing entries only for routes that are in active use
* At most one next-hop per destination maintained at each node
  * DSR may maintain several routes for a single destination

* Sequence numbers are used to avoid old/broken routes
* Sequence numbers prevent formation of routing loops

* Unused routes expire even if topology does not change

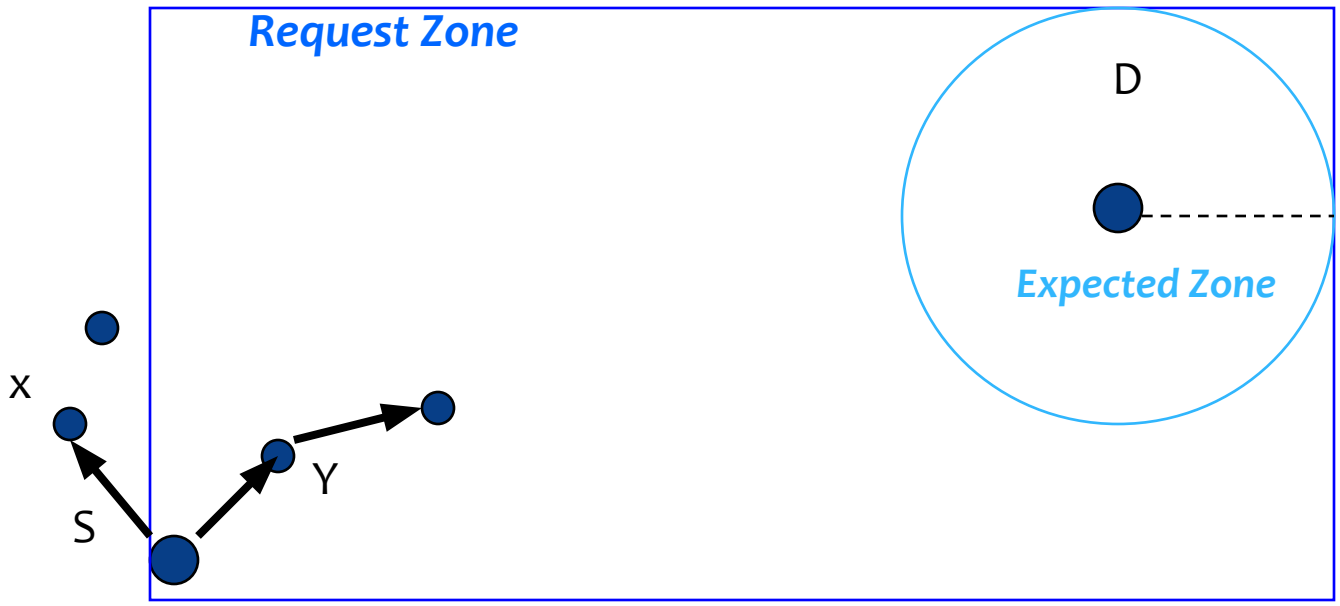
Location-Aided Routing (LAR)

* Exploits location information to limit scope of route request flooding
  * Location information may be obtained using GPS

* *Expected Zone* is determined as a region that is expected to hold the current location of the destination
  * Expected region determined based on potentially old location information, and knowledge of the destination's speed
* Route requests limited to a *Request Zone* that contains the Expected Zone and location of the sender node

Request Zone

* Define a Request Zone
* LAR is same as flooding, except that only nodes in request zone forward route request
* Smallest rectangle including S and expected zone for D



Location Aided Routing (LAR)

* Advantages
  * reduces the scope of route request flood
  * reduces overhead of route discovery
* Disadvantages
  * Nodes need to know their physical locations
  * Does not take into account possible existence of obstructions for radio transmissions

Desirable Attributes: Routing Algorithm

* Executes distributedly
* provides loop-free routes
* provides multiple routes
* Establishes routes quickly
* Minimizes communication overhead
* Selects optimal path

Tora Attributes

* On demand source initiated routing
* Nodes only maintain information about adjacent nodes (distributed)
* Provides multiple routes
* Creates loop-free routes
* Handles partitions by erasing routes

Motivation: TORA

* Observation: In hilly terrain, routing to a river's mouth is easy – just go downhill
* Idea: Turn network into hilly terrain
  * Different "landscape" for each destination
  * Assign "heights" to nodes such that when going downhill, destination is reached – in effect: orient edges between neighbors
  * **Necessary: resulting directed graph has to be cycle free**
  * **Route Creation**: establish sequence of directed links from source to destination. this is done by forming destination oriented directed acyclic graph (DAG)

Directed Acyclic Graph

* A graph G(V,E) is a directed graph with V nodes and E edges if every edge in the graph has an associated direction.
* A directed graph with no cycles is a DAG.
* A DAG is rooted at the destination if the destination is the only node with no downstream nodes, i.e., no links lead out of the destination. Such a DAG is often called a Destination oriented DAG.
* Creation of such a DAG from a source to a destination would contain multiple routes to the destination.

Reaction to topology changes

* **Route Maintenance:** Reaction to topology changes in order to reestablish routes
  * **When link is removed that was the last "outlet" of a node, reverse direction of all its other links (increase height!)**
  * **Reapply continuously, until each node except destination has at least a single outlet – will succeed in a connected graph!**

* **Route Erasure:** When a partition is detected in the network, all invalid routes must be removed from the network. This is done by making directed routes undirected (erasing routes).

TORA Specification

Node Height in TORA

* In TORA the height assigned to every node i is represented by a quintuple: **H[i] = (τ, oid, r, δ, i)**
* The first three fields represent the reference level.
  * τ: a time tag indicating the "time" of link failure.
  * oid: originator-id – unique id of the node that defined the new reference level.
  * r: reflection bit – used to divide each reference level into 2 sub-levels

TORA Specification

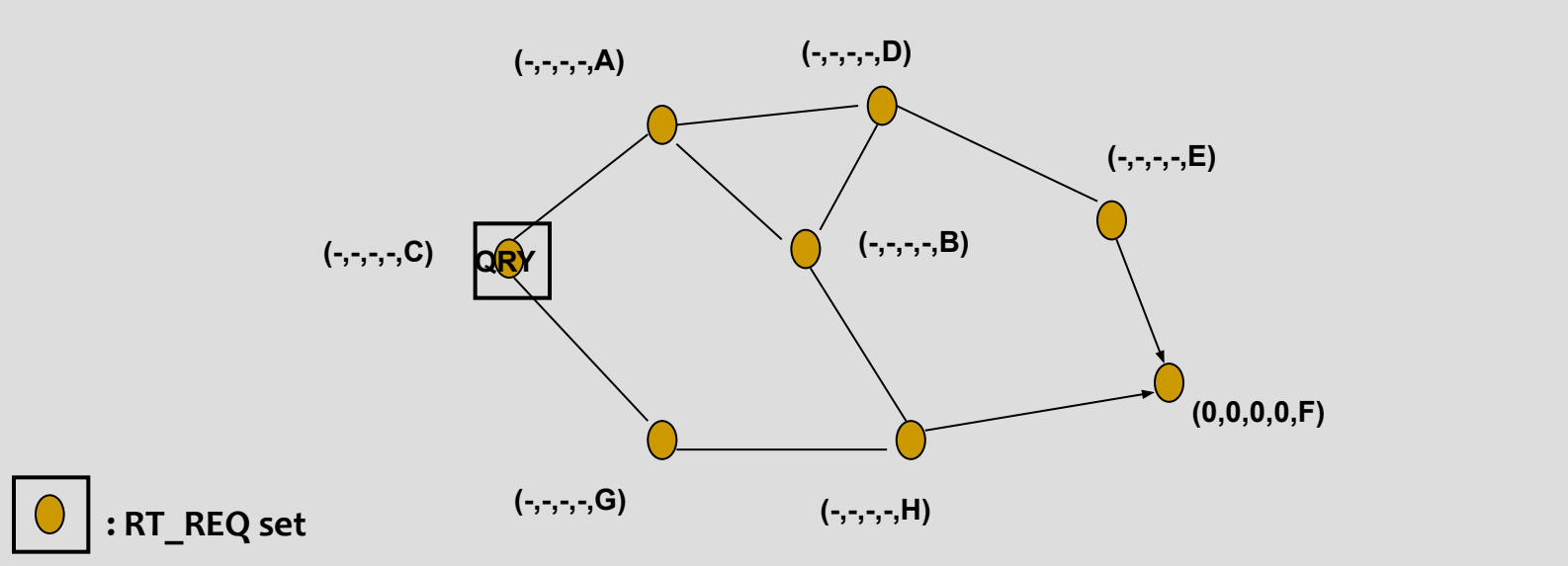
Node Height in TORA

- δ: integer used to order nodes with respect to unique reference level
- i: unique identifier of the node

* height of each node (except for the destination) is initially set to NULL: H[i] = (-, -, -, -, i)
* height of destination node d is initially set to ZERO: H[d] = (0, 0, 0, 0, d)
* each node maintains a height array containing the heights of neighbours: H[i][k]

Route Creation

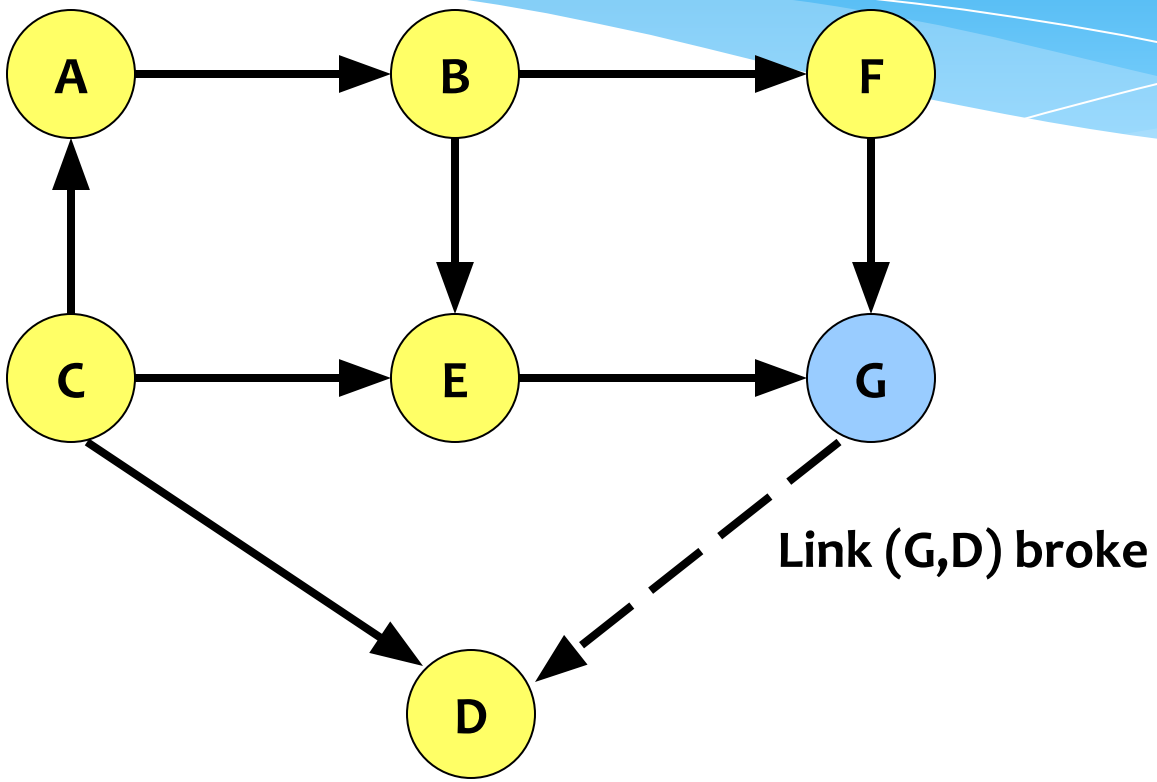
* node C requires route to F and broadcasts QRY packet



Temporally Ordered Routing Algorithm

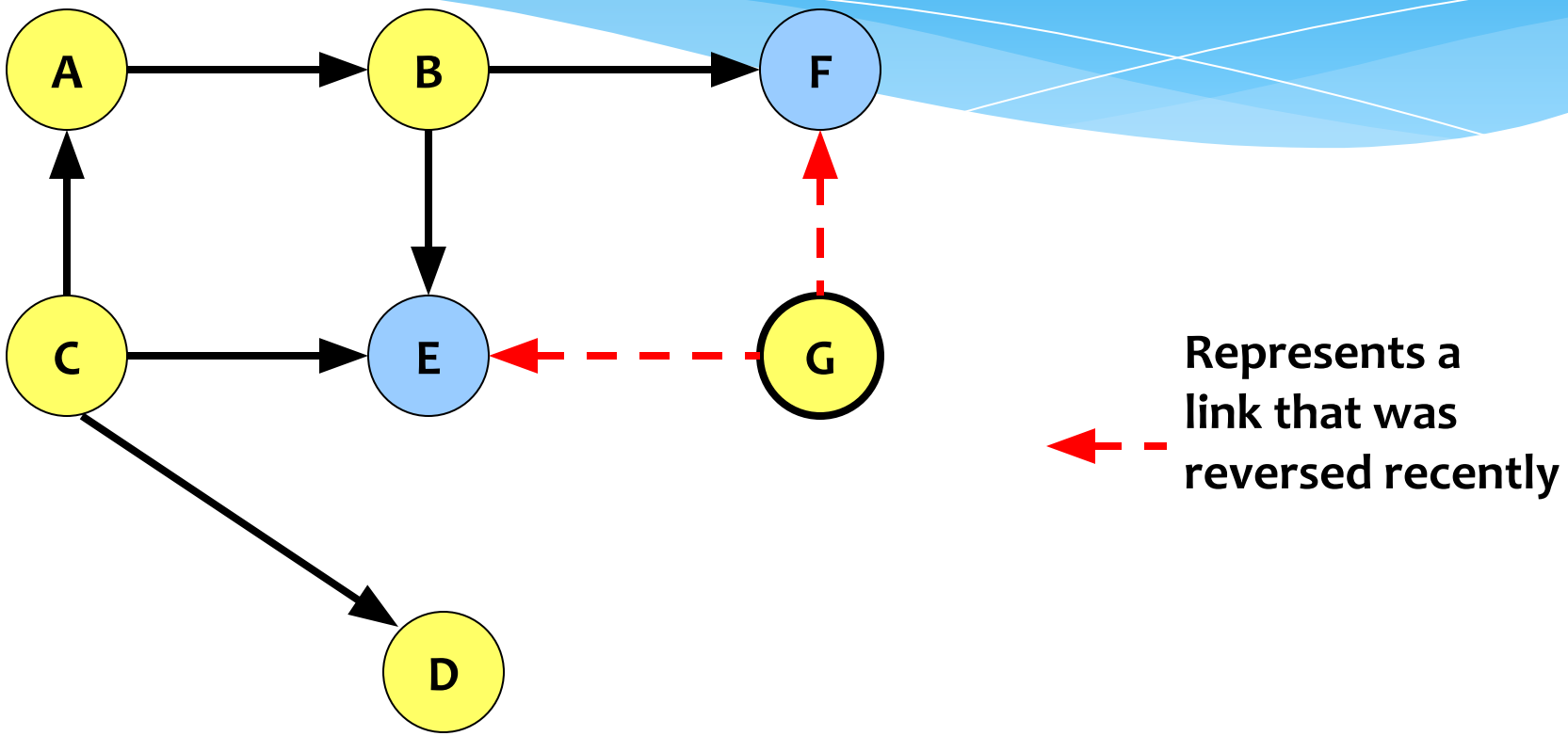
* Directed Acyclic Graph (DAG) rooted at destination is used to route packets
* Link Reversal algorithm used to update DAG (along with notion of "height")
* Algorithm is distributed and loop-free
* Recent result - Link reversal takes $O(n^2)$ time and message complexity to stabilize

TORA Example



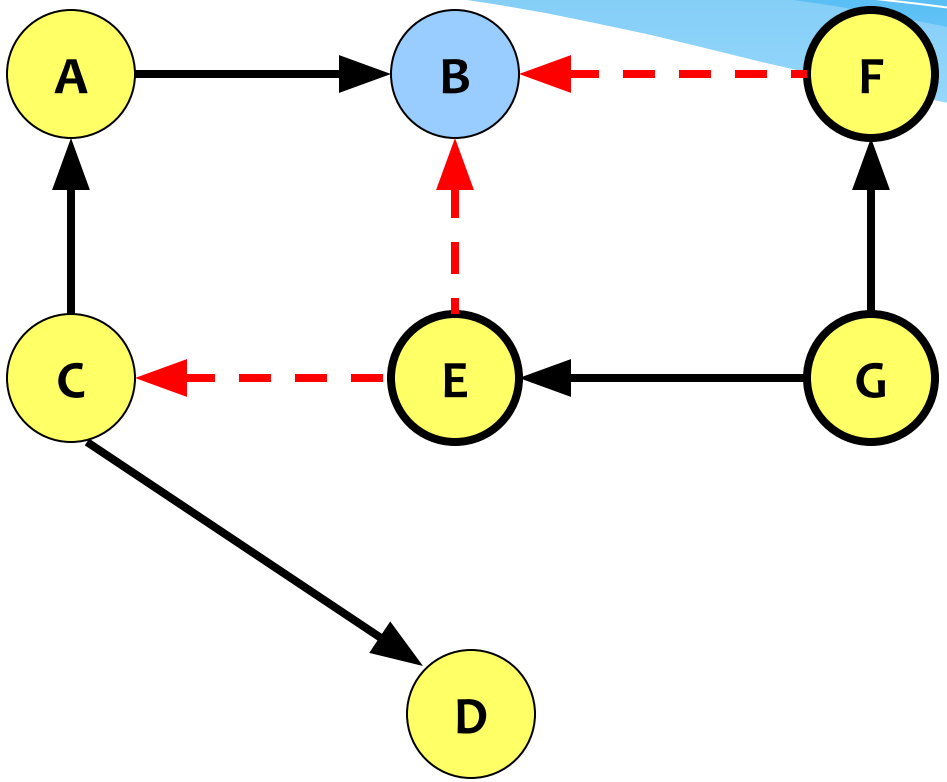
Node G has no outgoing links

TORA Example



Now nodes E and F have no outgoing links

TORA Example

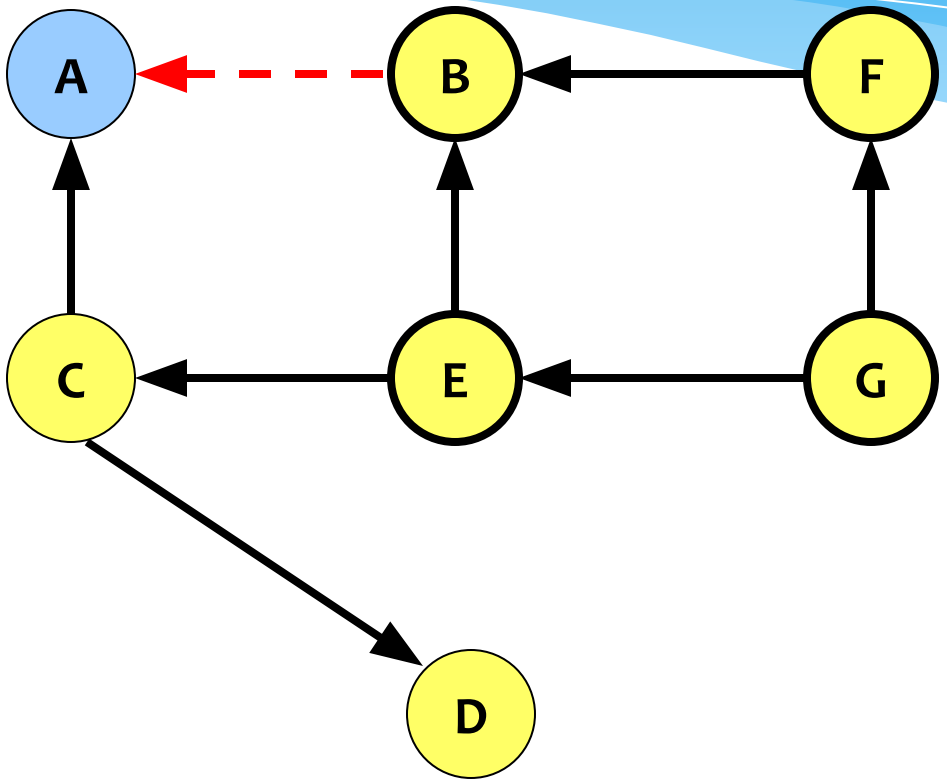


Represents a link that was reversed recently

Nodes E and F *do not* reverse links from node G

Now node B has no outgoing links

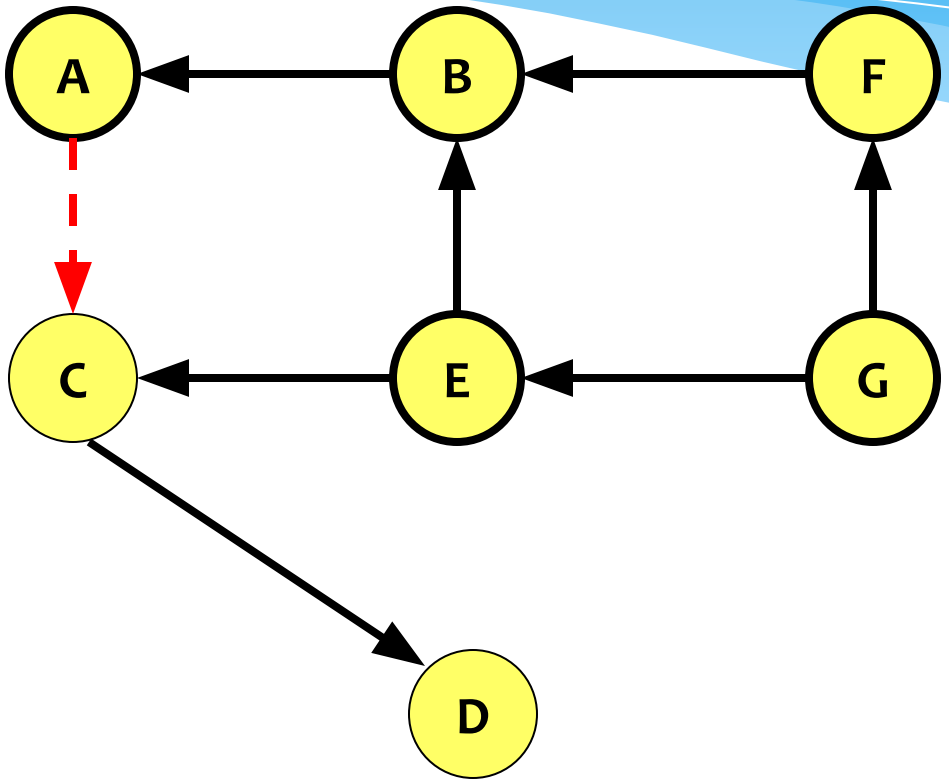
TORA Example



Represents a link that was reversed recently

Now node A has no outgoing links

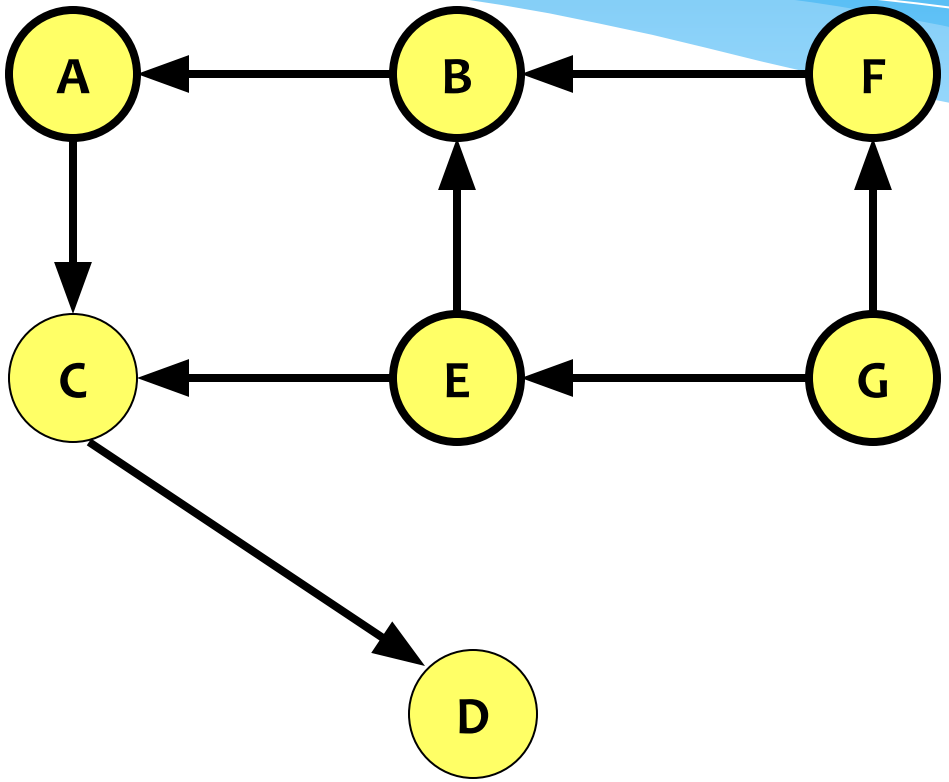
TORA Example



Represents a link that was reversed recently

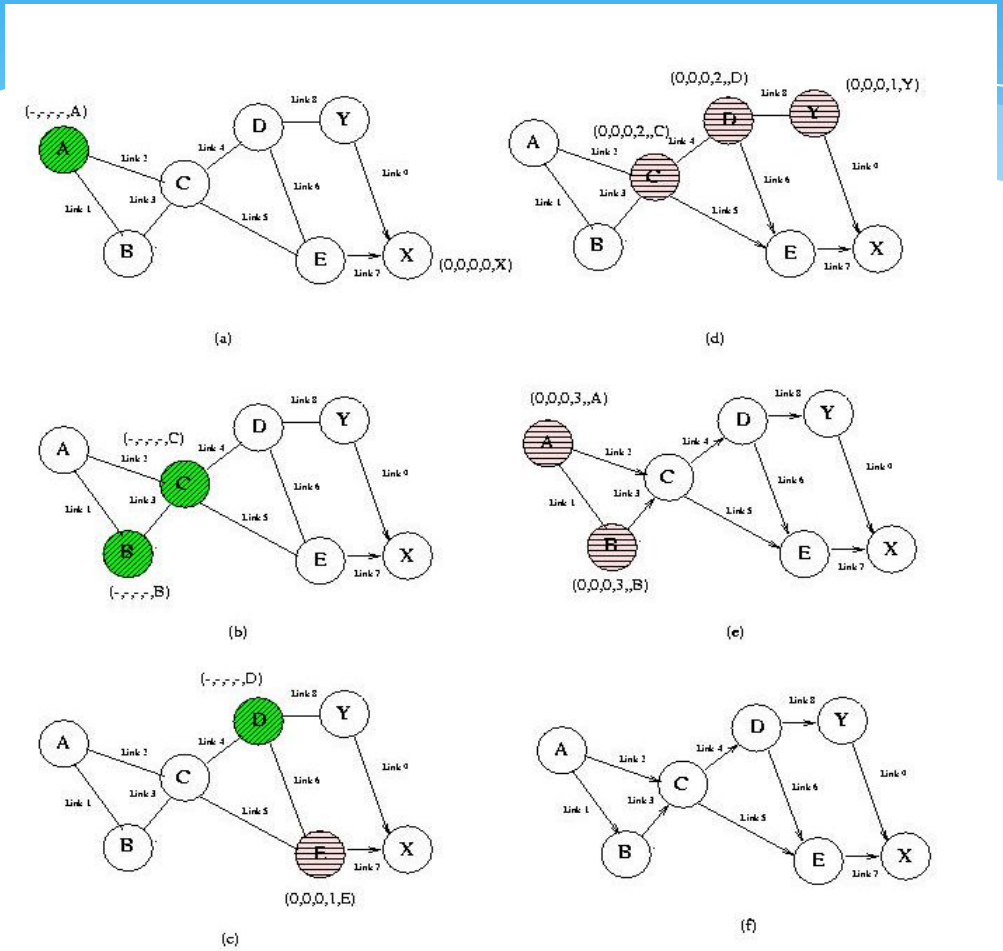
Now all nodes (except destination D) have outgoing links

TORA Example



DAG has been restored with only the destination as a sink

TORA: Example



- The destination is X.
- A needs to find a route to X.
- A sends a QRY packet to C.
- C and B broadcast this message.
- E knows that X is one hop away.
- D broadcasts QRY message (contains height which is currently NULL).
- What does E do ?

Other routing protocols

* Geographic Routing Protocols
  * Location Aided Routing (LAR)
  * Distance Routing Effect Algorithm for Mobility (DREAM)
  * Greedy Perimeter Stateless Routing (GPSR)

* Hybrid Routing Protocols
  * Zone Routing Protocol (ZRP)

Clusterhead Gateway Switch Routing (CGSR)

* Nodes organized into hierarchy of clusters.
* Each node has a clusterhead, selected using an election.
* Nodes send packet through clusterheads.
* Clusterheads communicate amongst themselves using DSDV.
  * Two clusters are connected through a gateway node

Wireless Routing Protocol (WRP)

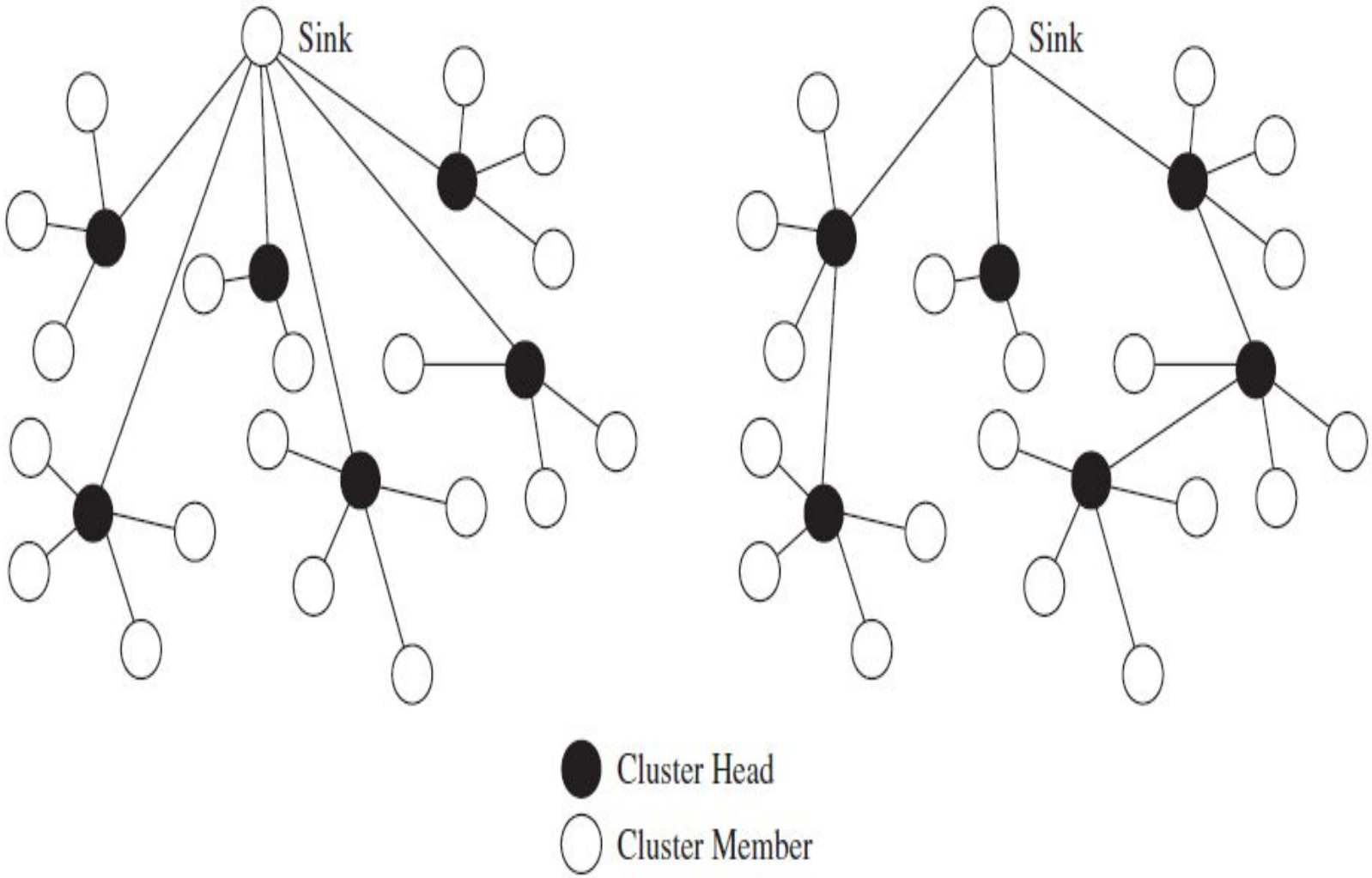
* Maintains multiple tables
  * Distance, routing, link-cost, etc.
* Link change messages exchanged only between neighbors
* Loop freedom using novel algorithm
  * Uses predecessor hop information

Other Table-Driven Protocols

* Optimized Link State Routing Protocol (OLSR) – RFC 3626
  * Optimization of link-state routing to wireless

* Topology Dissemination Based on Reverse Path Forwarding (TBRPF) - RFC 3684
  * Also based on link-state routing

* Scalability & Efficiency
* Hierarchical solutions may reduce collisions in the wireless medium and facilitate the duty cycling of sensor nodes for increased energy efficiency.
* Clustering may also facilitate the routing process, but may lead to longer routes than many flat routing protocols.
* Clustering also facilitates in-network aggregation of sensor data, because data coming from co-located sensors (which may monitor overlapping regions of the environment) are likely to pass through the same cluster head.


Sink
Sink
Cluster Head
Cluster Member

The Problem

* The reason we need network protocol such as LEACH is due to the fact that a node in the network is no longer useful when its battery dies.

* This protocol allows us to space out the lifespan of the nodes, allowing it to do only the minimum work it needs to transmit data

LEACH

* Hierarchical clustering algorithm is the Low-Energy Adaptive Clustering Hierarchy that combines a clustering approach with MAC-layer techniques.

* LEACH assumes that every cluster head can directly communicate with the base station.
* Cluster heads are responsible for all communication between their cluster members and a base station and the aggregation of data coming from its cluster members in order to eliminate redundancies.

LEACH

* LEACH can achieve significant energy savings (depending on how much redundancy can be removed) and sensor nodes (apart from the cluster heads) are not responsible for forwarding other nodes' data.

LEACH's Two Phases

* The LEACH network has two phases: the set-up phase and the steady-state
  * The Set-Up Phase
    * Where cluster-heads are chosen
  * The Steady-State
    * The cluster-head is maintained
    * When data is transmitted between nodes

Stochastic Threshold Algorithm

- Cluster-heads can be chosen *stochastically* (randomly based) on this algorithm:

$$T(n) = \frac{P}{1 - P \times (r \bmod P^{-1})} \qquad \forall n \in G$$

$$T(n) = 0 \qquad \forall n \notin G$$

Where n is a random number between 0 and 1
P is the cluster-head probability and
G is the set of nodes that weren't cluster-heads the previous rounds

- If n < T(n), then that node becomes a cluster-head
- The algorithm is designed so that each node becomes a cluster-head at least once

Deterministic Threshold Algorithm

* A modified version of this protocol is known as LEACH-C (or LEACH Centralized)
* This version has a *deterministic* threshold algorithm, which takes into account the amount of energy in the node…

$$T(n)_{new} = \frac{P}{1 - P \times (r \bmod P^{-1})} \frac{E_{n_current}}{E_{n_max}}$$

Where $E_{n_current}$ is the current amount of energy and

E_{n_max} is the initial amount of energy

Deterministic Threshold Algorithm

* …and/or whether or not the node was recently a

$$T(n)_{new} = \frac{P}{1 - P \times (r \bmod P^{-1})} \left[\frac{E_{n_current}}{E_{n_max}} + (r_s \operatorname{div} P^{-1}) \left(1 - \frac{E_{n_current}}{E_{n_max}} \right) \right]$$

LEACH: Operation

* Periodic process
* Three phases per round:
  * Advertisement
    * Election and membership
  * Setup
    * Schedule creation
  * Steady-State
    * Data transmission

LEACH: Advertisement

* Cluster head self-election
  * Status advertised broadcast to nearby nodes
* Non-cluster heads must listen to the medium
  * Choose membership based on signal strength
    * RSSI
    * E_b/N_o

LEACH: Setup

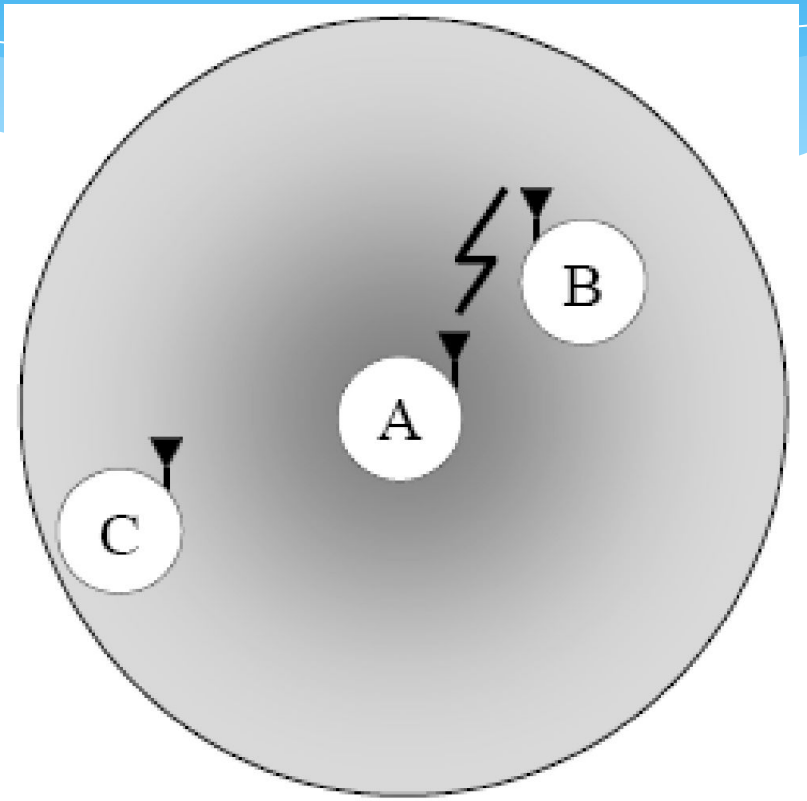
* Nodes broadcast membership status
  * CSMA
* Cluster heads must listen to the medium
* TDMA schedule created
  * Dynamic number of time slices

LEACH: Data Transmission

* Nodes sleep until time slice
* Cluster heads must listen to each slice
* Cluster heads aggregate/compress and transmit once per cycle
* Phase continues until the end of the round
  * Time determined *a priori*

LEACH: Interference Avoidance

* TDMA intra-cluster
* CDMA inter-cluster
  * Spreading codes determined randomly
    * Non-overlapping modulation may be NP-Complete
  * Broadcast during advertisement phase



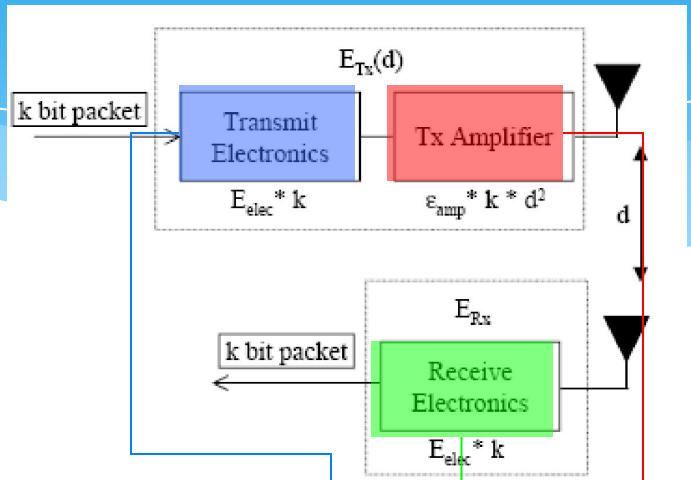
What's the Difference?

* REMEMBER: The goal of these protocol is to increase the life of the network

* The changes between the LEACH stochastic algorithm and the LEACH-C deterministic algorithm alone is proven to increase the FND (First Node Dies) lifetime by 30% and the HND (Half Node Dies) lifetime by 20%

Radio Model

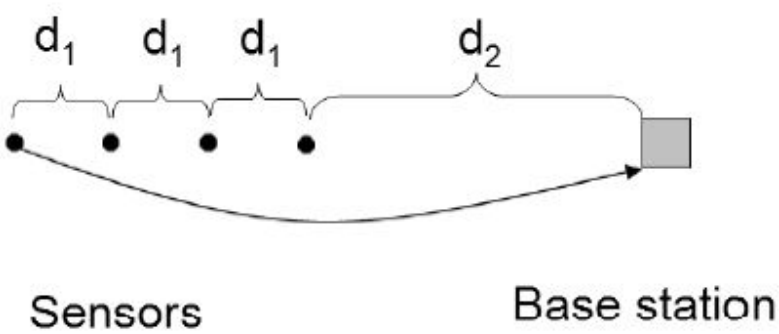
* Designed around acceptable E_b/N_o
* E_{elec} = 50nJ/bit
  * Energy dissipation for transmit and receive
* ε_{amp} = 100pJ/bit/m²
  * Energy dissipation for transmit amplifier
* k = Packet size
* d = Distance



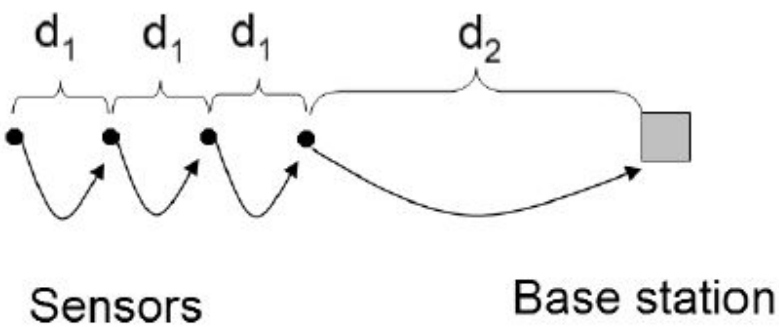
$$E_{Tx}(k,d) = (E_{elec} * k) + (\varepsilon_{amp} * k * d^2)$$

$$E_{Rx}(k) = E_{elec} * k$$

Direct v. Minimum Transmission



(a) Direct transmission



(b) Minimum transmission energy

* The amount of energy used in figure (a) can be modeled by this formula:
  * $\varepsilon_{amp} k(3d_1 + d_2)^2$
* Whereas the amount of energy used in figure (b) uses this formula:
  * $\varepsilon_{amp} k(3d_1^2 + d_2^2)$

The Amount of Energy Depletion

* This is the formula for the amount of energy depletion by data transfer:

The energy being dissipated to run the transmitter: $E_{elec} = 50$ nJ/bit

The energy dissipation of the transmission amplifier: $\varepsilon_{amp} = 100$ pJ/bit/m2

Transmission costs: $E_{Tx}(k,d) = E_{elec}\, k + \varepsilon_{amp}\, kd^{\lambda}$

Receiving costs: $E_{Rx}(k) = E_{elec}\, k$

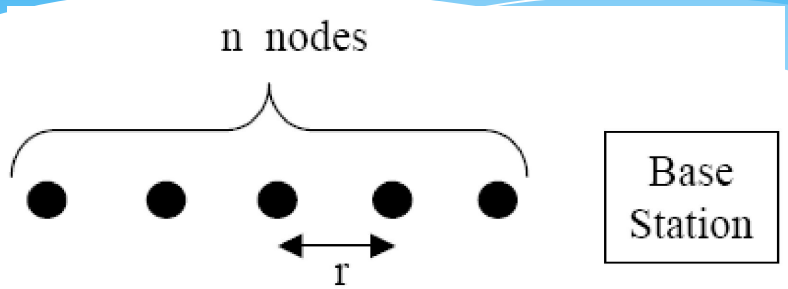
Where k is the length of the message in bits

d is distance between nodes, and

λ represents the path-loss exponent ($\lambda \geq 2$).

MTE vs Direct-Transmission

When is Direct-Transmission Better?



$$E_{direct} = k(E_{elec} + \varepsilon_{amp} n^2 r^2)$$

$$E_{MTE} = k((2n-1)E_{elec} + \varepsilon_{amp} n r^2)$$

For MTE, a node at distance *nr* requires *n* transmits of distance *r*, and *n*-1 receives

$E_{direct} < E_{MTE}$ when:

$$\frac{E_{elec}}{\varepsilon_{amp}} > \frac{r^2 n}{2}$$

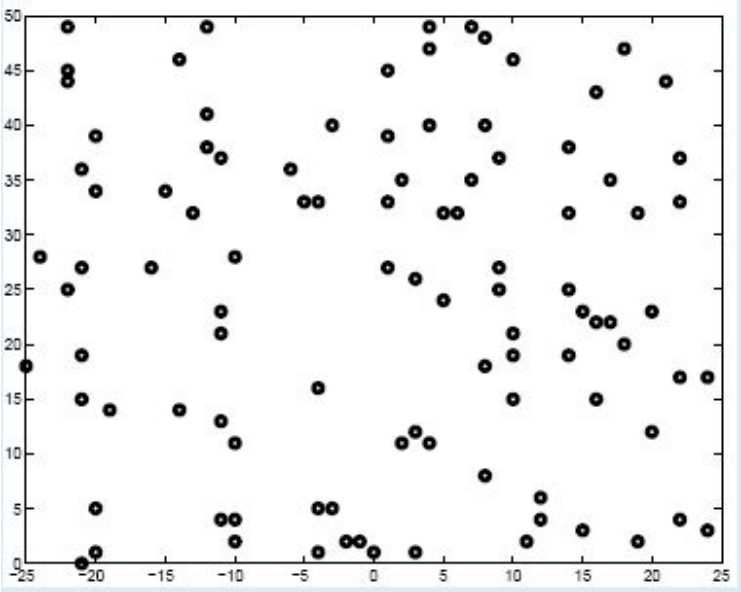
- High radio operation costs favor direct-transmission
- Low transmit amplifier costs (i.e. distance to the sink) favor direct transmission
- Small inter-node distances favor MTE

Existing Routing Protocols

* LEACH is compared against three other routing protocols:
  * Direct-Transmission
    * Single-hop
  * Minimum-Transmission Energy
    * Multi-hop
  * Static Clustering
    * Multi-hop

Performance: Parameters

* MATLAB Simulator
* 100-node random network
* E_{elec} = 50nj/bit
* ε_{amp} = 100pJ/bit/m2
* k = 2000 bits



Performance: System Lifetime

* Experiments repeated for different maximum energy levels
* LEACH gains:
  * 8x life expectancy for first node
  * 3x life expectancy for last node

Energy (J/node)	Protocol	Round first node dies	Round last node dies
0.25	Direct	55	117
	MTE	5	221
	Static Clustering	41	67
	LEACH	394	665
0.5	Direct	109	234
	MTE	8	429
	Static Clustering	80	110
	LEACH	932	1312
1	Direct	217	468
	MTE	15	843
	Static Clustering	106	240
	LEACH	1848	2608

Conclusions

* LEACH is completely distributed
  * No centralized control system
* LEACH outperforms:
  * Direct-Transmission in most cases
  * MTE in many cases
  * Static clustering in effectively all cases
* LEACH can reduce communication costs by up to 8x
* LEACH keeps the first node alive for up to 8x longer and the last node by up to 3x longer